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Impact of health literacy and its interventions on health outcomes in those with atrial fibrillation: a systematic review protocol

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Impact of health literacy and its interventions on health outcomes in those with atrial fibrillation: a systematic review protocol

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Keywords: Atrial fibrillation, precision health, health literacy, health outcomes, quality of life

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ABSTRACT

Introduction

Atrial fibrillation (AF) is associated with increased risk of stroke, heart failure and death. Health literacy, an aspect that falls within precision health, has been recognised as an important factor. We will be focusing on the impact of these interventions specifically to AF and its health outcomes.

Methods and Analysis

This protocol is informed by the Preferred Reporting Items for Systematic review and Meta-Analysis Protocols (PRISMA-P). The results will be reported in line with the Preferred Reporting Items for Systematic review and Meta-Analysis (PRISMA) to determine the impacts of health literacy interventions on atrial fibrillation outcomes. Searches will be carried out on databases including MEDLINE, EMBASE, Web of Science, CINAHL, Emcare, Cochrane library, and Google Scholar. Citations will be collected via Endnote 20, then into Covidence for duplicate removal, and article screening. Extraction will occur using a standardised extraction tool and studies will be synthesised using best evidence synthesis. Downs and Black's checklist will be used for risk of bias and assessment of overall quality of evidence will utilise the Grading of Recommendations, Assessment, Development, and Evaluation approach (GRADE).

Ethics and Dissemination

Approval from human research ethics committee is not required. Dissemination will occur in peer-reviewed journals and conference presentations.

STRENGTHS AND LIMITATIONS OF STUDY

- This review will add to efforts operationalising precision health by extending this specifically to health literacy interventions.
- The review will adhere to PRISMA guidelines specific to intervention studies, use the Downs and Black checklist for quality appraisal and the GRADE recommendations for certainty of evidence.
- The meta-synthesis is anticipated to highlight gaps in the use of health literacy interventions as a form of precision health on health outcomes in patients with AF.
- Different study designs and definitions of health literacy interventions may limit the extent of comparative analysis.

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INTRODUCTION

Background & Rationale

Atrial fibrillation is the most common sustained cardiac arrhythmia and represents significant morbidity and mortality worldwide.(1, 2) Epidemiologic information extracted from the Global Health Data Exchange database in 2020 showed a worldwide prevalence of 37,575 million cases (0.51% of population). The prevalence of AF has increased over the last 20 years and is predicted to continue to increase.(3, 4) In addition to an increased risk of strokes, heart failure, and all-cause death, atrial fibrillation continues to contribute to the growing burden on the healthcare system and its resources.(1)

The management of AF is complex and requires consideration to a number of factors such as stroke prophylaxis, rate and rhythm control and risk factor management.(5-10) This opens avenues for utilization of precision health to improve patient outcomes.(11, 12) Derived from the phrase 'precision medicine', which was itself developed from 'personalised medicine' and 'stratified medicine',(13, 14) precision health has had a significant presence in public health, as well as specialties like clinical oncology, where the study of genomics, genome variations, and their implications in individuals have been heavily researched.(14, 15) This is seen in previous definitions for precision health, such as involving only 'variations within a patient's genetic profile as a means to optimise disease management'.(16) However, the definition of precision health in a broader sense has been expanded to include a range of additional concepts. These include the processes of lifetime monitoring and collection of health data, analysis of collected data, and subsequent informed, personalised decision making and interventions.(11-14) Viana et al's(12) scoping review identified nine types of data that can be collected, which includes sociodemographic, anthropometric, clinical, genomic, perspective, behavioural, psychological, nutritional, and socioenvironmental data. Furthermore, the resulting treatment and health outcomes post-intervention can be integrated back into this cycle, providing feedback for further sophistication of interventions, and improvement of health outcomes.(11, 13) A created precision health system actively involves all parties associated with the patient's care, emphasising a cooperative shared decision making model between patient and healthcare providers within a multidisciplinary fashion.(11, 17-19) Even still, despite definitions involving the concepts mentioned above, the exact definition of precision health remains uncertain, with no consensus yet reached.(13, 20)

Our review will be focusing on one aspect within precision health – health literacy. In defining health literacy, Sørensen et al.(21) reviewed nineteen publications and seventeen explicit definitions, and concluded it as involving people's knowledge and interactions with health information via four dimensions: accessing, understanding, appraising, and applying. These dimensions are then situated within 3 domains: as a patient within the healthcare system, disease prevention, and health promotion within the wider community. Sørensen et al.(21) emphasises that the health literacy ultimately involves people's ability to judge decisions concerning their quality of life. Similarly, Nutbeam(22) discusses 3 types of health literacy: functional health literacy encompasses basic skills in reading and writing as to perform the role of patient within the healthcare system; interactive health literacy describes more advanced knowledge within cognitive, literacy, and social skills that allow people to extract and derive meaning from different communication forms; and lastly, critical health literacy allows patients to critically analyse information and subsequently have greater control

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over their own life. Health literacy plays a critical role within a patient's healthcare and quality of life, and improving it has been a recognised goal by the World Health Organisation (WHO).(23, 24)

Objectives

This systematic review will explore precision health, in particular, the impact of health literacy and its interventions, on atrial fibrillation outcomes. Thus, the systematic review will endeavour to answer the following:

1. Precision health initiatives have generally focused on behavioural and sociodemographic/clinical data. How does the addition of health literacy interventions generate additional benefits?
2. What questionnaires and tools are currently available that have been used to assess health literacy both generally and in AF?
3. What initiatives have been used to improve health literacy, and how effective are these initiatives in improving both knowledge and subsequent beneficial actions for AF?
4. Is there any correlation between health literacy (both in terms of knowledge of AF and subsequent actions), quality of life and health outcomes in AF?

METHODS

Guidelines and Registration

This protocol for our systematic review was done with guidance and adherence to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 guidelines,(25, 26), Cochrane guidelines, (27) and other reviews already conducted in this area. Registration was submitted to the International Prospective Register of Systematic Reviews (PROSPERO), an international database for systematic reviews prospectively registered by the Centre for Reviews and Dissemination of the University of York (https://www.crd.york.ac.uk/prospero). This occurred on [29th April 2022] and was last updated on [10th May 2022] (registration number [CRD42022304835]).

Patient and Public Involvement

This systematic review will involve only existing reviews, and thus, will not involve patients or members of the public throughout any stage of the review.

Eligibility Criteria

Selection of studies will consider the following:

Study Designs

Randomised control trials (RCTs) and observational studies will be included. Conference presentations, thesis, secondary research (review articles) will be excluded.

Population

Included studies that have enrolled individuals with a primary diagnosis of non-valvular AF. Any age, baseline physical activity, or different backgrounds will be considered, as well as those in various settings, such as hospital-based, hospital outpatient clinicals, private clinics, primary care, research-

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based specialist university clinics. No restrictions will be placed on duration of symptoms nor time since diagnosis. Studies in which participants are diagnosed with valvular AF will be excluded.

Interventions

This review will consider studies that evaluate interventions and initiatives focused on health literacy. Interventions may endeavour to improve any dimension,(21) typology,(22) or similar aspect of health literacy, such as a patient's ability to read, write, approach numeracy, communicate, and/or use electronic technology.(28) Interventions considered can address health literacy in any domain(21) or setting, and may be in the form of different mediums, which can include, but is not limited to, verbal communication, written communication, visual aids, and supportive systems.(28, 29) We also acknowledge the growing usage of digital initiatives, such as online resources, web-based resources, social media, and mobile applications.

This intervention fits within Gambhir et al.'s(11) model of precision health within the "Integrated health portal", allowing for learning and adaptation that feeds back into the rest of the model.

Comparator

In addition to usual clinical care, we will be considering behavioural data, sociodemographic data, and clinical data as part of our comparators. Viana et al.'s(12) scoping review on precision health research showed that these were the top three types of data used for precision health (behavioural (48%), sociodemographic (44%), clinical (43.6%)).

Outcomes

Three categories of outcomes will be collected: outcomes concerned with the knowledge and understanding of AF following intervention; outcomes measuring any resulting patient action following intervention; and finally, outcomes measuring health, wellness, and quality of life. Due to the broad nature of measures for outcomes, the following is not an exhaustive list. Any additional outcomes suitable for the three categories listed above will be considered and agreed upon by all researchers involved.

Examples of outcomes include

- Those related to knowledge and understanding
    - Test of Functional Health Literacy in Adults (TOFHLA)(30)/short-TOFHLA (s-TOFHLA)(31)
    - Rapid Estimate of Adult Literacy in Medicine (REALM)(32)/REALM-Short Form (REALM-SF)(33)
    - Medical achievement reading test (MART)(34)
- Those related to action
    - Self-reported measures
- Those related to health, wellness, and quality of life
    - All-cause mortality
    - Hospitalisations, and re-hospitalisation rates
    - Heart function
    - Quality of life(35)

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- Generic instruments – Medical Outcome Study Short-Form 36 (SF-36),(36) 12-Item Short-Form Health Survey (SF-12),(37) EuroQoL 5-Dimensional (EQ-5D)(38, 39)
- AF-specific instruments – Atrial Fibrillation Effect on QualiTy-of-Life (AFEQT),(40) AF-QOL-18,(41) QLAF questionnaire,(42) Atrial Fibrillation Quality of Life Questionnaire (AFQLQ)(43)

- Symptom scales(35)
- Physical activity – Internal Physical Activity Questionnaire (IPAQ),(44) Lawton-Brody iADL scale,(45) Katz-ADL scale(46)
- Self-reported measures
  - Ability to perform ADLs
  - Work and productivity

Information Sources

Searches will be carried out on MEDLINE (via PubMed platform), EMBASE (via OVID platform), Web of Science (via Clarivate platform), CINAHL (via EBSCOhost platform), Emcare (via OVID platform), Cochrane Library, and Google Scholar.(47)

Additionally, to ensure completeness, gray literature(48) will be included and citation searching of included articles will be undertaken.

Search Strategy

The draft search strategy was developed initially using the PubMed platform, medical subject headings (MeSH) and key phrases identified during preliminary searches. These were initially performed in January 2022 and then updated in April 2022 and assisted in refining search terms within the PICO structure. Phrases specific to the intervention, comparator, and outcome components were developed by first identifying core key phrases relevant to our questions, then searching PubMed using these phrases and expanding on synonyms using existing articles and search strategies. We based our search on previous literature including Viana et al.'s(12) review of precision health. There will be no limitations on date or language; any studies in languages other than English will be translated using Google Translate.

The draft search strategy was reviewed by a librarian specialising in medicine and was also reviewed by the research team. A complete search strategy, which has been adapted to syntax and subject headings of other databases, is included in Appendix 1.

Data Management & Selection Process

Citations from databases will be imported to EndNote 20 (Clarivate Analytics) for article collection, then to Covidence for the remaining processes. Duplicates will be removed electronically via Covidence and checked manually. Eligibility for inclusion will involve two independent reviewers, and firstly will be done based on title and abstract screening, followed by full-text review. At both stages, reviewers will be blinded to each other's decisions until discussions occur. Any disagreements at either stage will be resolved through discussion, and/or with a third reviewer. Decisions will be recorded using Covidence.

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Data Extraction

Data will be extracted from studies by two independent reviewers using piloting forms and a standardised extraction tool. Extraction will be structured by the following domains:

1. Article characteristics, which include title, name of first author, publication year, country of study, study type;
2. Participant details, which include number of participants, demographics, characteristics, study location, recruitment procedures, inclusion and exclusion criteria;
3. Methods, which include identification of AF, definition and application of precision health initiatives, exposure measurement; and,
4. Outcomes, which include measurement tool, timing of measurements, method of aggregation, summary data.

Authors of papers will be contacted to request missing or additional data, where required.

Data Synthesis

Best evidence synthesis will be used. Meta-analysis of intervention (effectiveness) can be carried out; however, we will need to account for substantial heterogeneity between and within included studies. Specifically for our review, outcomes will need to be compared according to the specific health literacy intervention carried out. The process for meta-analysis will be informed following data extraction, based on the assumptions of homogeneity.

If homogeneity is sufficient between groups within these categories, a consensus will be reached between all authors regarding the inclusion within the meta-analysis either as a large group or several subgroups. If eligible studies are clearly homogenous, studies will be allocated to appropriate sub-groups for analysis. This process will be done by each reviewer independently, determining the factors that would allow for the most accurate comparisons to be made.

If a broad consensus is reached for meta-analysis grouping, the process will continue. Failing this, supplementary discussion will occur. If further progress cannot be made, all authors will discuss suitability of meta-analysis or meta-synthesis.

To assess the impact of heterogeneity across studies, a statistic test of inconsistency will be carried out, returning I^2 which will be reported as a percentage. The Cochrane Handbook for Systematic Reviews of Interventions(27) provides a suggested guide for interpretation of I^2, group percentages into 0-40%, 30-60%, 50-90%, 75-100% to potentially mean no importance, moderate, substantial, and considerable heterogeneity respectively. If present, heterogeneity will be explored, and potential options, such as excluding studies with obvious outlying results, will be explored.

Risk of Bias & Confidence in Cumulative Evidence

Strategies to combat potential publication data will include searches of unpublished studies, contact authors in their respective field, and including grey literature. Further statistical tests, such as the Begg and Mazumbar's rank correlation test and Egger's linear regression model, may be applied to each category and overall analyses. On publication bias being detected, Duval and Tweedie's trim and fill

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correction may be applied, and the resultant effect sizes and 95% confidence intervals examined in further detail.

The Downs and Black's 27 item checklist(49) to assess methodological quality of randomised and non-randomised trials. It provides: an overall score for study quality, and a profile of scores for the quality of reporting internal validity, power and external validity.

A Summary of Findings (SoF) will be created using GRADEpro GDT. The SoF will present the following information where appropriate: absolute risks for the treatment and control, estimates of relative risk, and a ranking of the quality of the evidence based on the risk of bias, directness, heterogeneity, precision and risk of publication bias of the review results. The GRADE(50) approach will be utilised to assess quality of evidence.

CONTRIBUTORSHIP STATEMENT

All authors have contributed to the focus of this systematic review topic and revisions associated with this protocol manuscript. Specifically, DK, TL, and RM were responsible for the initial conceptualisation of this work. TL, DK, RM led the planning and design, and all authors were involved in the conduct and reporting of this manuscript. All authors made substantial contributions to the drafting and critical revision of the work and all authors approved the final manuscript.

COMPETING INTERESTS

There are no competing interests.

FINANCIAL SUPPORT

Dr Mahajan has served on the advisory board of Abbott and Medtronic. The University of Adelaide reports receiving on behalf of Dr Mahajan lecture and/or consulting fees from Abbott, Bayer, Biotronik, Medtronic, and Pfizer. The University of Adelaide reports receiving on behalf of Dr Mahajan research funding from Abbott, Bayer and Medtronic. The other authors do not report any conflicts of interest with regards to this research.

DATA SHARING

Data sharing not applicable as no datasets generated and/or analysed for this study.

ETHICS AND DISSEMINATION

Approval from a human research ethics committee is not required as human participants will not be involved in this review. We do intend to publish this review on completion in a high-quality peer-reviewed journals, conference presentations and internal staff research meetings.

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APPENDIX

Appendix 1 – Search Strategy

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Logic Grids as of 21/04/2022

MEDLINE (via Pubmed platform)

Atrial Fibrillation	**Health Literacy Initiatives**	**Outcomes**
96,171 hits	*585,959 hits*	*4,482,653 hits*
"atrial fibrillation"[mh] OR "atrial fibrillation"[tiab] OR "auricular fibrillation"[tiab]	"health literacy"[mh] OR "patient medication knowledge"[mh] OR "health communication"[mh] OR "health promotion"[mh] OR "patient education as topic"[mh] OR "health education"[mh] OR "patient participation"[mh] OR "telemedicine"[mh] OR "internet-based intervention"[mh] OR "social media"[mh] OR "mobile applications"[mh] OR "smartphone"[mh] OR "educational technology"[mh] OR "health literacy"[tiab] OR "patient medication knowledge"[tiab] OR "health education"[tiab] OR "health communication*"[tiab] OR "health promotion*"[tiab] OR "patient participation"[tiab] OR "telemedicine"[tiab] OR "telehealth"[tiab] OR "internet-based intervention*"[tiab] OR "social media"[tiab] OR "mobile applications"[tiab] OR "smartphone"[tiab] OR "educational technolog*"[tiab] OR "health literacy interventions"[tiab] OR "education program*"[tiab] OR "educational program*"[tiab] OR "wellness program*"[tiab] OR "health campaign*"[tiab] OR "patient education"[tiab] OR "patient empowerment"[tiab] OR "patient engagement"[tiab] OR "patient activation"[tiab] OR "mobile health"[tiab] OR "mhealth"[tiab] OR "ehealth"[tiab] OR "interactive health literacy"[tiab] OR "web-based intervention*"[tiab] OR "web based intervention*"[tiab] OR "online intervention*"[tiab] OR "internet intervention*"[tiab] OR "mobile app*"[tiab] OR "smartphone app*"[tiab] OR "digital health"[tiab] OR "instructional technolog*"[tiab] OR (("technolog*"[tiab] OR "computer"[tiab] OR "internet"[tiab] OR "online"[tiab]) AND "health literacy"[tiab])	"health knowledge, attitudes, practice"[mh] OR "patient preference"[mh] OR "treatment adherence and compliance"[mh] OR "prognosis"[mh] OR "patient reported outcome measures"[mh] OR "patient outcome assessment"[mh] OR "quality of life"[mh] OR "health status"[mh] OR "activities of daily living"[mh] OR "mortality"[mh] OR "morbidity"[mh] OR "hospitalization"[mh] OR "health knowledge"[tiab] OR "health attitudes"[tiab] OR "health practice"[tiab] OR "patient preference*"[tiab] OR "treatment adherence and compliance"[tiab] OR "prognosis"[tiab] OR "patient reported outcome measure*"[tiab] OR "patient outcome assessment*"[tiab] OR "quality of life"[tiab] OR "health status"[tiab] OR "activities of daily living"[tiab] OR "mortality"[tiab] OR "morbidity"[tiab] OR "hospitalization*"[tiab] OR "hospitalisation*"[tiab] OR "health literacy questionnaire"[tiab] OR "health understanding"[tiab] OR "health competenc*"[tiab] OR "behaviour change"[tiab] OR "treatment adherence"[tiab] OR "therapeutic adherence"[tiab] OR "health-related quality of life"[tiab] OR "HRQOL"[tiab] OR "ADL"[tiab] OR "functional capacity"[tiab] OR "stroke risk"[tiab] OR "atrial fibrillation complication*"[tiab] OR "atrial fibrillation burden"[tiab]
		647 combined hits

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Logic Grids as of 21/04/2022
EMBASE (via OVID platform)

Atrial Fibrillation	Health Literacy Initiatives	Outcomes
172,878 hits	*640,603 hits*	*4,596,175 hits*
exp "atrial fibrillation"/ or "atrial fibrillation".ti,ab.	"health literacy"/ or "patient education"/ or "health education"/ or "medical information"/ or "health promotion"/ or "patient participation"/ or "patient engagement"/ or "telemedicine"/ or "telehealth"/ or "web-based intervention"/ or "social media"/ or exp "mobile application"/ or exp "smartphone"/ or "educational technology"/ or "health literacy".ti,ab. or "patient education".ti,ab. or "health education".ti,ab. or "medical information".ti,ab. or "health promotion".ti,ab. or "patient participation".ti,ab. or "patient engagement".ti,ab. or "telemedicine".ti,ab. or "telehealth".ti,ab. or "web?based intervention*".ti,ab. or "social media".ti,ab. or "mobile app*".ti,ab. or "smartphone".ti,ab. or "educational technolog*".ti,ab. or "patient medication knowledge".ti,ab. or "health communication*".ti,ab. or "internet-based intervention*".ti,ab. or "health literacy interventions".ti,ab. or "education* program*".ti,ab. or "wellness program*".ti,ab. or "health campaign*".ti,ab. or "patient empowerment".ti,ab. or "patient activation".ti,ab. or "mobile health".ti,ab. or "mhealth".ti,ab. or "ehealth".ti,ab. or "interactive health literacy".ti,ab. or "online intervention*".ti,ab. or "internet intervention*".ti,ab. or "smartphone app*".ti,ab. or "digital health".ti,ab. or "instructional technolog*".ti,ab. or (("technolog*" or "computer" or "internet" or "online").ti,ab. AND "health literacy".ti,ab.)	"patient preference"/ or exp "patient compliance"/ or "prognosis"/ or "patient-reported outcome"/ or "outcome assessment"/ or exp "quality of life"/ or "health status"/ or "fitness"/ or "functional status"/ or "health disparity"/ or "physical mobility"/ or "daily life activity"/ or "mortality"/ or "morbidity"/ or "hospitalization/" or "behaviour change"/ or "patient preference*".ti,ab. or "patient compliance".ti,ab. or "prognosis".ti,ab. or "patient-reported outcome*".ti,ab. or "outcome assessment".ti,ab. or "quality of life".ti,ab. or "health status".ti,ab. or "fitness".ti,ab. or "functional status".ti,ab. or "health disparity".ti,ab. or "physical mobility".ti,ab. or "daily life activity".ti,ab. or "activities of daily living".ti,ab. or "mortality".ti,ab. or "morbidity".ti,ab. or "hospitali?ation".ti,ab. or "behaviour change".ti,ab. or "health knowledge".ti,ab. or "health attitudes".ti,ab. or "health practice".ti,ab. or "health literacy questionnaire".ti,ab. or "health understanding".ti,ab. or "health competenc*".ti,ab. or "treatment adherence".ti,ab. or "therapeutic adherence".ti,ab. or "health-related quality of life".ti,ab. or "HRQOL".ti,ab. or "ADL".ti,ab. or "stroke risk".ti,ab. or "atrial fibrillation complication*".ti,ab. or "atrial fibrillation burden".ti,ab.
		1,141 combined hits

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Logic Grids as of 21/04/2022

Web of Science (via Clarivate platform)

Atrial Fibrillation	**Health Literacy Initiatives**	**Outcomes**
114,231 hits	*405,908 hits*	*2,449,416 hits*
TS=("atrial fibrillation")	TS=("health literacy" OR "patient medication knowledge" OR "health communication" OR "health promotion" OR "patient education" OR "health education" OR "patient participation" OR "telemedicine" OR "internet-based intervention" OR "social media" OR "mobile applications" OR "smartphone" OR "educational technology" OR "health communication*" OR "health promotion*" OR "health literacy interventions" OR "education* program*" OR "wellness program*" OR "health campaign*" OR "patient education" OR "patient empowerment" OR "patient engagement" OR "patient activation" OR "mobile health" OR "mhealth" OR "ehealth" OR "interactive health literacy" OR "web-based intervention*" OR "web based intervention*" OR "online intervention*" OR "internet intervention*" OR "mobile app*" OR "smartphone app*" OR "digital health" OR "instructional technolog*" OR (("technolog*" OR "computer" OR "internet" OR "online") AND "health literacy"))	TS=("health knowledge" OR "health attitudes" OR "health practice" OR "patient preference" OR "treatment adherence" OR "treatment compliance" OR "prognosis" OR "patient reported outcome measures" OR "patient outcome assessment" OR "quality of life" OR "health status" OR "activities of daily living" OR "mortality" OR "morbidity" OR "hospitali?ation*" OR "health literacy questionnaire" OR "health understanding" OR "health competenc*" OR "behaviour change" OR "treatment adherence" OR "therapeutic adherence" OR "health-related quality of life" OR "HRQOL" OR "ADL" OR "functional capacity" OR "stroke risk" OR "atrial fibrillation complication*" OR "atrial fibrillation burden")
		319 combined hits

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Logic Grids as of 21/04/2022
CINAHL (via EBSCOhost platform)

Atrial Fibrillation	**Health Literacy Initiatives**	**Outcomes**
39,320 hits	*298,598 hits*	*1,150,418 hits*
MH "atrial fibrillation" OR TI "atrial fibrillation" OR AB "atrial fibrillation"	MH "health literacy" OR MH "health promotion" OR MH "health education" OR MH "patient education" OR MH "telemedicine" OR MH "telehealth" OR MH "internet-based intervention" OR MH "social media+" OR MH "mobile applications" OR MH "smartphone" OR MH "educational technology" OR TI ("health literacy" OR "health promotion*" OR "health education" OR "patient education" OR "telemedicine" OR "telehealth" OR "internet-based intervention" OR "social media" OR "mobile app*" OR "smartphone" OR "educational technolog*" OR "patient medication knowledge" OR "health communication" OR "patient participation" OR "health literacy interventions" OR "educational program*" OR "education program*" OR "wellness program*" OR "health campaign*" OR "patient empowerment" OR "patient engagement" OR "patient activation" OR "mobile health" OR "mhealth" OR "ehealth" OR "smartphone app*" OR "interactive health literacy" OR "web-based intervention*" OR "web based intervention*" OR "online intervention*" OR "internet intervention*" OR "digital health" OR "instructional technolog*" OR (("technolog*" OR "computer" OR "internet" OR "online") AND "health literacy"))) OR AB ("health literacy" OR "health promotion*" OR "health education" OR "patient education" OR "telemedicine" OR "telehealth" OR "internet-based intervention" OR "social media" OR "mobile app*" OR "smartphone" OR "educational technolog*" OR "patient medication knowledge" OR "health communication" OR "patient participation" OR "health literacy interventions" OR "educational program*" OR "education program*" OR "wellness program*" OR "health campaign*" OR "patient empowerment" OR "patient engagement" OR "patient activation" OR "mobile health" OR "mhealth" OR "ehealth" OR "smartphone app*" OR "interactive health literacy" OR "web-based intervention*" OR "web based intervention*" OR "online intervention*" OR "internet intervention*" OR "digital health" OR "instructional technolog*" OR (("technolog*" OR "computer" OR "internet" OR "online") AND "health literacy")))	MH "health knowledge" OR MH "patient preference" OR MH "prognosis" OR MH "treatment outcomes" OR MH "outcome assessment" OR MH "patient-reported outcomes" OR MH "quality of life" OR MH "health status+" OR MH "activities of daily living" OR MH "mortality" OR MH "morbidity" OR MH "hospitalization" OR MH "behavioural changes" OR MH "functional status" OR TI ("health knowledge" OR "health practice" OR "patient preference" OR "prognosis" OR "treatment outcomes" OR "outcome assessment" OR "patient-reported outcomes" OR "quality of life" OR "health status" OR "activities of daily living" OR "mortality" OR "morbidity" OR "hospitali#ation" OR "behavioural changes" OR "functional status" OR "treatment adherence" OR "therapeutic adherence" OR "health literacy questionnaire" OR "health understanding" OR "health competenc*" OR "health-related quality of life" OR "HRQOL" OR "ADL" OR "stroke risk" OR "atrial fibrillation complication*" OR "atrial fibrillation burden") OR AB ("health knowledge" OR "health practice" OR "patient preference" OR "prognosis" OR "treatment outcomes" OR "outcome assessment" OR "patient-reported outcomes" OR "quality of life" OR "health status" OR "activities of daily living" OR "mortality" OR "morbidity" OR "hospitali#ation" OR "behavioural changes" OR "functional status" OR "treatment adherence" OR "therapeutic adherence" OR "health literacy questionnaire" OR "health understanding" OR "health competenc*" OR "health-related quality of life" OR "HRQOL" OR "ADL" OR "stroke risk" OR "atrial fibrillation complication*" OR "atrial fibrillation burden")
		232 combined hits

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Logic Grids as of 21/04/2022

Emcare (via OVID platform)

Atrial Fibrillation	**Health Literacy Initiatives**	**Outcomes**
41,599 hits	*276,531 hits*	*1,210,113 hits*
exp "atrial fibrillation"/ or "atrial fibrillation".ti,ab.	"health literacy"/ or "patient education"/ or "health education"/ or "medical information"/ or "health promotion"/ or "patient participation"/ or "patient engagement"/ or "telemedicine"/ or "telehealth"/ or "web-based intervention"/ or "social media"/ or exp "mobile application"/ or exp "smartphone"/ or "educational technology"/ or "health literacy".ti,ab. or "patient education".ti,ab. or "health education".ti,ab. or "medical information".ti,ab. or "health promotion".ti,ab. or "patient participation".ti,ab. or "patient engagement".ti,ab. or "telemedicine".ti,ab. or "telehealth".ti,ab. or "web?based intervention*".ti,ab. or "social media".ti,ab. or "mobile app*".ti,ab. or "smartphone".ti,ab. or "educational technolog*".ti,ab. or "patient medication knowledge".ti,ab. or "health communication*".ti,ab. or "internet-based intervention*".ti,ab. or "health literacy interventions".ti,ab. or "education* program*".ti,ab. or "wellness program*".ti,ab. or "health campaign*".ti,ab. or "patient empowerment".ti,ab. or "patient activation".ti,ab. or "mobile health".ti,ab. or "mhealth".ti,ab. or "ehealth".ti,ab. or "interactive health literacy".ti,ab. or "online intervention*".ti,ab. or "internet intervention*".ti,ab. or "smartphone app*".ti,ab. or "digital health".ti,ab. or "instructional technolog*".ti,ab. or (("technolog*" or "computer" or "internet" or "online").ti,ab. AND "health literacy".ti,ab.)	"patient preference"/ or exp "patient compliance"/ or "prognosis"/ or "patient-reported outcome"/ or "outcome assessment"/ or exp "quality of life"/ or "health status"/ or "fitness"/ or "functional status"/ or "health disparity"/ or "physical mobility"/ or "daily life activity"/ or "mortality"/ or "morbidity"/ or "hospitalization/" or "behaviour change"/ or "patient preference*".ti,ab. or "patient compliance".ti,ab. or "prognosis".ti,ab. or "patient-reported outcome*".ti,ab. or "outcome assessment".ti,ab. or "quality of life".ti,ab. or "health status".ti,ab. or "fitness".ti,ab. or "functional status".ti,ab. or "health disparity".ti,ab. or "physical mobility".ti,ab. or "daily life activity".ti,ab. or "activities of daily living".ti,ab. or "mortality".ti,ab. or "morbidity".ti,ab. or "hospitali?ation".ti,ab. or "behaviour change".ti,ab. or "health knowledge".ti,ab. or "health attitudes".ti,ab. or "health practice".ti,ab. or "health literacy questionnaire".ti,ab. or "health understanding".ti,ab. or "health competenc*".ti,ab. or "treatment adherence".ti,ab. or "therapeutic adherence".ti,ab. or "health-related quality of life".ti,ab. or "HRQOL".ti,ab. or "ADL".ti,ab. or "stroke risk".ti,ab. or "atrial fibrillation complication*".ti,ab. or "atrial fibrillation burden".ti,ab.
		337 combined hits

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Logic Grids as of 21/04/2022
Cochrane Library

Atrial Fibrillation	Health Literacy Initiatives	Outcomes
13,040 hits	*63,570 hits*	*428,751 hits*
[mh "atrial fibrillation"] OR "atrial fibrillation":ti,ab OR "auricular fibrillation":ti,ab	[mh "health literacy"] OR [mh "patient medication knowledge"] OR [mh "health communication"] OR [mh "health promotion"] OR [mh "patient education as topic"] OR [mh "health education"] OR [mh "patient participation"] OR [mh "telemedicine"] OR [mh "internet-based intervention"] OR [mh "social media"] OR [mh "mobile applications"] OR [mh "smartphone"] OR [mh "educational technology"] OR "health literacy":ti,ab OR "patient medication knowledge":ti,ab OR "health education":ti,ab OR (health NEXT communication*):ti,ab OR (health NEXT promotion*):ti,ab OR "patient participation":ti,ab OR "telemedicine":ti,ab OR "telehealth":ti,ab OR (internet-based NEXT intervention*):ti,ab OR "social media":ti,ab OR "mobile applications":ti,ab OR "smartphone":ti,ab OR (educational NEXT technolog*):ti,ab OR "health literacy interventions":ti,ab OR (education NEXT program*):ti,ab OR (educational program*):ti,ab OR (wellness NEXT program*):ti,ab OR (health NEXT campaign*):ti,ab OR "patient education":ti,ab OR "patient empowerment":ti,ab OR "patient engagement":ti,ab OR "patient activation":ti,ab OR "mobile health":ti,ab OR "mhealth":ti,ab OR "ehealth":ti,ab OR "interactive health literacy":ti,ab OR (web-based NEXT intervention*):ti,ab OR (web NEXT based NEXT intervention*):ti,ab OR (online NEXT intervention*):ti,ab OR (internet NEXT intervention*):ti,ab OR (mobile NEXT app*):ti,ab OR (smartphone NEXT app*):ti,ab OR "digital health":ti,ab OR (instructional NEXT technolog*):ti,ab OR ((technolog*:ti,ab OR computer:ti,ab OR internet:ti,ab OR online:ti,ab) AND "health literacy":ti,ab)	[mh "health knowledge, attitudes, practice"] OR [mh "patient preference"] OR [mh "treatment adherence and compliance"] OR [mh "prognosis"] OR [mh "patient reported outcome measures"] OR [mh "patient outcome assessment"] OR [mh "quality of life"] OR [mh "health status"] OR [mh "activities of daily living"] OR [mh "mortality"] OR [mh "morbidity"] OR [mh "hospitalization"] OR "health knowledge":ti,ab OR "health attitudes":ti,ab OR "health practice":ti,ab OR "patient preference*":ti,ab OR "treatment adherence and compliance":ti,ab OR "prognosis":ti,ab OR (patient NEXT reported NEXT outcome NEXT measure*):ti,ab OR (patient NEXT outcome NEXT assessment*):ti,ab OR "quality of life":ti,ab OR "health status":ti,ab OR "activities of daily living":ti,ab OR "mortality":ti,ab OR "morbidity":ti,ab OR (hospitalization*):ti,ab OR (hospitalisation*):ti,ab OR "health literacy questionnaire":ti,ab OR "health understanding":ti,ab OR (health NEXT competenc*):ti,ab OR "behaviour change":ti,ab OR "treatment adherence":ti,ab OR "therapeutic adherence":ti,ab OR "health-related quality of life":ti,ab OR "HRQOL":ti,ab OR "ADL":ti,ab OR "functional capacity":ti,ab OR "stroke risk":ti,ab OR "atrial fibrillation complication*":ti,ab OR "atrial fibrillation burden":ti,ab
		174 combined hits

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Location in Manuscript
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	Page 1: Title, Page 2: Abstract
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	Page 4: Abstract (CRD42022304835)
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Page 1: Title Page
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Page 1: Title Page, Page 8: Contributorship Statement
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	Page 8, Financial Disclosures
Sponsor	5b	Provide name for the review funder and/or sponsor	N/A
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	N/A
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Page 3
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Page 4
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting,	Page 4-6

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		time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Page 6
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Appendix 1
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Page 6
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	Page 6
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Page 7
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	Page 7
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Page 5-6
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Page 7-8
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	Page 7
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication	Page 7

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		bias across studies, selective reporting within studies)	
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	Page 8

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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BMJ Open

Impact of health literacy and its interventions on health outcomes in those with atrial fibrillation: a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2022-065407.R1
Article Type:	Protocol
Date Submitted by the Author:	18-Oct-2022
Complete List of Authors:	Lathlean, Timothy; The University of Adelaide Adelaide Medical School, Faculty of Health and Medical Sciences; South Australian Health and Medical Research Institute Limited, Kieu, Don; The University of Adelaide Adelaide Medical School, Faculty of Health and Medical Sciences Franke, Kyle; The University of Adelaide Adelaide Medical School, Faculty of Health and Medical Sciences O'Callaghan, Nathan; Commonwealth Scientific and Industrial Research Organisation, Precision Health Future Science Platform; South Australian Health and Medical Research Institute Limited Boyd, Mark A; The University of Adelaide Adelaide Medical School, Faculty of Health and Medical Sciences; Northern Adelaide Local Health Network, Lyell McEwin Hospital Mahajan, Rajiv; The University of Adelaide Faculty of Health and Medical Sciences, Adelaide Medical School; Northern Adelaide Local Health Network, Lyell McEwin Hospital
Primary Subject Heading:	Cardiovascular medicine
Secondary Subject Heading:	Patient-centred medicine
Keywords:	Health informatics < BIOTECHNOLOGY & BIOINFORMATICS, Adult cardiology < CARDIOLOGY, PREVENTIVE MEDICINE

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Manuscripts

Impact of health literacy and its interventions on health outcomes in those with atrial fibrillation: a systematic review protocol

AUTHORS AND AFFILIATIONS

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CONFLICT OF INTEREST DISCLOSURES:

The University of Adelaide reports receiving on behalf of Dr Mahajan lecture and/or consulting fees from Abbott, Medtronic, Bayer, Biotronik, and Pfizer. The University of Adelaide reports receiving on behalf of Dr Mahajan research funding from Abbott, Bayer, and Medtronic. The remaining authors do not report any conflicts of interest.

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Keywords: Atrial fibrillation, precision health, health literacy, health outcomes, quality of life

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ABSTRACT

Introduction

Atrial fibrillation (AF) is associated with increased risk of stroke, heart failure and death. Health literacy, an aspect that falls within precision health, has been recognised as an important factor. We will be focusing on the impact of these interventions specifically to AF and its health outcomes.

Methods and Analysis

This protocol is informed by the Preferred Reporting Items for Systematic review and Meta-Analysis Protocols (PRISMA-P). The results will be reported in line with the Preferred Reporting Items for Systematic review and Meta-Analysis (PRISMA) to determine the impacts of health literacy interventions on atrial fibrillation outcomes. Searches will be carried out on databases including MEDLINE, EMBASE, Web of Science, CINAHL, Emcare, Cochrane library, and Google Scholar. Citations will be collected via Endnote 20, then into Covidence for duplicate removal, and article screening. Extraction will occur using a standardised extraction tool and studies will be synthesised using best evidence synthesis. Downs and Black's checklist will be used for risk of bias and assessment of overall quality of evidence will utilise the Grading of Recommendations, Assessment, Development, and Evaluation approach (GRADE).

Ethics and Dissemination

Approval from human research ethics committee is not required. Dissemination will occur in peer-reviewed journals and conference presentations.

STRENGTHS AND LIMITATIONS OF STUDY

- This will be the first review to operationalise health literacy interventions in people with atrial fibrillation.
- The protocol for this systematic review and meta-analysis follows the PRISMA-P guidelines.
- This systematic review will use the Downs and Black checklist for quality appraisal and the GRADE recommendations for certainty of evidence.
- The meta-analysis will provide a synthesis of the evidence of health literacy interventions consistent with a) knowledge b) action and c) health outcomes.
- Different study designs and definitions of health literacy interventions as well as varied outcomes may limit the ability the extent of comparative analysis.

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INTRODUCTION

Background & Rationale

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia and represents significant morbidity and mortality worldwide.(1) The worldwide prevalence of AF is estimated at 0.51% of population and projected to increase further.(2) In addition to an increased risk of stroke, heart failure, and all-cause death, AF continues to contribute to the growing burden on the healthcare system and its resources.(1)

Health literacy plays a critical role within a patient's healthcare and quality of life, and improving it has been a recognised goal by the World Health Organisation (WHO).(3, 4) Health literacy has been defined according to three types: functional health literacy, interactive health literacy and critical health literacy. (5) It involves people's knowledge and interactions with health information via four dimensions: accessing, understanding, appraising, and applying. (6) Health literacy can be considered as an aspect of 'precision health', which involve the processes of lifetime monitoring and collection of health data, analysis of collected data, and subsequent informed, personalised decision making and interventions.(7-10) A recent scoping review (9) identified nine categories of data that can be collected: sociodemographic, anthropometric, clinical, genomic, perspective, behavioural, psychological, nutritional, and socioenvironmental data.

Currently, a range of definitions and terms have been used in the health literacy literature, with different aims and contexts, which has created uncertainty and a lack of clarity as to what these constitute. This is particularly pronounced when considering health literacy interventions as a core component of precision health initiatives. Our systematic review and meta-analysis aims to evaluate the effectiveness of health literacy interventions for people with AF. In doing so, we aim to synthesise the current literature to identify the impacts of health literacy interventions on a) health literacy and knowledge; b) patient action and behaviour change arising from health literacy and knowledge and c) health outcomes (e.g. quality of life, hospitalisations and/or morbidity/mortality) as a consequence to action taken arising from improvements in health literacy.

Objectives

This systematic review will explore precision health, in particular, the impact of health literacy and its interventions, on atrial fibrillation outcomes. Thus, the systematic review will endeavour to answer the following:

1. How do health literacy interventions generate additional benefits for precision health initiatives?
2. What questionnaires and tools are currently available and have been used to assess health literacy interventions generally and in AF?
3. How effective are health literacy interventions in improving both knowledge and subsequent beneficial actions for AF?
4. How effective are health literacy interventions that improve knowledge and subsequent beneficial actions in resulting in improved health outcomes such as quality of life, hospitalisations and reduced morbidity/mortality?

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METHODS

Guidelines and Registration

This protocol for our systematic review was done with guidance and adherence to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 guidelines,(11, 12), Cochrane guidelines, (13) and other reviews already conducted in this area.(14, 15) Registration was submitted to the International Prospective Register of Systematic Reviews (PROSPERO), an international database for systematic reviews prospectively registered by the Centre for Reviews and Dissemination of the University of York (https://www.crd.york.ac.uk/prospero). This occurred on [29th April 2022] and was last updated on [10th May 2022] (registration number [CRD42022304835]).

Patient and Public Involvement

This systematic review will involve only existing primary research papers.

Eligibility Criteria

Selection of studies will consider the following:

Study Designs

Randomised control trials (RCTs) and observational studies will be included. Conference presentations, thesis, secondary research (review articles) will be excluded.

Population

Included studies that have enrolled individuals with a primary diagnosis of non-valvular AF. Any age, baseline physical activity, or different backgrounds will be considered, as well as those in various settings, such as hospital-based, hospital outpatient clinicals, private clinics, primary care, research-based specialist university clinics. No restrictions will be placed on duration of symptoms nor time since diagnosis. Studies in which participants are diagnosed with valvular AF will be excluded.

Interventions

This review will consider studies that evaluate interventions and initiatives focused on health literacy that are aimed at improving any of the following 1) knowledge and/or literacy 2) patient action and behaviour change and 3) health outcomes, including quality of life, hospitalisations, morbidity/mortality.

Interventions may endeavour to improve any dimension,(6) typology,(5) or similar aspect of health literacy, such as a patient's ability to read, write, approach numeracy, communicate, and/or use electronic technology.(16)

Interventions considered can address health literacy in any domain(6) or setting, and may be in the form of different medium. This can involve face-to-face mediums, such as verbal communication, written communication, visual aids, and supportive systems.(16, 17). We also acknowledge the growing usage of remote mediums and digital initiatives, such as online resources, web-based resources, social media, and mobile applications.

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Such intervention fits within Gambhir et al.'s(7) model of precision health within the "Integrated health portal", allowing for learning and adaptation that feeds back into the rest of the model.

Comparator

In addition to usual clinical care, we will be considering behavioural data, sociodemographic data, and clinical data as part of our comparators. Viana et al.'s(9) scoping review on precision health research showed that these were the top three types of data used for precision health (behavioural (48%), sociodemographic (44%), clinical (43.6%)).

Outcomes

Three categories of outcomes will be collected: outcomes concerned with the knowledge and understanding of AF following intervention; outcomes measuring any resulting patient action following intervention; and finally, outcomes measuring health, wellness, and quality of life. Due to the broad nature of measures for outcomes, the following is not an exhaustive list. Any additional outcomes suitable for the three categories listed above will be considered and agreed upon by all researchers involved.

Examples of outcomes include

- Those related to knowledge and understanding
    - Test of Functional Health Literacy in Adults (TOFHLA)(18)/short-TOFHLA (s-TOFHLA)(19)
    - Rapid Estimate of Adult Literacy in Medicine (REALM)(20)/REALM-Short Form (REALM-SF)(21)
    - Medical achievement reading test (MART)(22)
- Those related to action
    - Self-reported measures
- Those related to health, wellness, and quality of life
    - All-cause mortality
    - Hospitalisations, and re-hospitalisation rates
    - Heart function
    - Quality of life(23)
        - Generic instruments – Medical Outcome Study Short-Form 36 (SF-36),(24) 12-Item Short-Form Health Survey (SF-12),(25) EuroQoL 5-Dimensional (EQ-5D)(26, 27)
        - AF-specific instruments – Atrial Fibrillation Effect on QualiTy-of-Life (AFEQT),(28) AF-QOL-18,(29) QLAF questionnaire,(30) Atrial Fibrillation Quality of Life Questionnaire (AFQLQ)(31)
    - Symptom scales(23)
    - Physical activity – Internal Physical Activity Questionnaire (IPAQ),(32) Lawton-Brody iADL scale,(33) Katz-ADL scale(34)
    - Self-reported measures
        - Ability to perform ADLs
        - Work and productivity

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Information Sources

Searches will be carried out on MEDLINE (via PubMed platform), EMBASE (via OVID platform), Web of Science (via Clarivate platform), CINAHL (via EBSCOhost platform), Emcare (via OVID platform), Cochrane Library, and Google Scholar.(35)

Additionally, to ensure completeness, gray literature(36) will be included and citation searching of included articles will be undertaken. Gray literature, or evidence not published in commercial publications can include academic papers, such as theses/dissertations, research and/or governmental reports, conference papers. We intend to search for such literature through resources such as OpenGrey, Scopus, and Proquest Digital Dissertations and Digital Dissertations and Conference Papers Index (CPI). We also aim to hand-search through relevant conference proceedings and abstracts such as the Cardiac Society of Australia and New Zealand (CSANZ), Asia Pacific Heart Rhythm Society (APHRS) and Heart Rhythm Society (HRS) Conferences.

Search Strategy

The draft search strategy was developed initially using the PubMed platform, medical subject headings (MeSH) and key phrases identified during preliminary searches. These were initially performed in January 2022 and then updated in April 2022 and assisted in refining search terms within the PICO structure. Phrases specific to the intervention, comparator, and outcome components were developed by first identifying core key phrases relevant to our questions, then searching PubMed using these phrases and expanding on synonyms using existing articles and search strategies. We based our search on previous literature including Viana et al.'s(9) review of precision health. The search will be limited to 2012 onwards as well as English language.

The draft search strategy was reviewed by a librarian specialising in medicine and was also reviewed by the research team. A complete search strategy, which has been adapted to syntax and subject headings of other databases, is included in Appendix 1.

Data Management & Selection Process

Citations from databases will be imported to EndNote 20 (Clarivate Analytics) for article collection, then to Covidence for the remaining processes. Duplicates will be removed electronically via Covidence and checked manually. Eligibility for inclusion will involve two independent reviewers, and firstly will be done based on title and abstract screening, followed by full-text review. At both stages, reviewers will be blinded to each other's decisions until discussions occur.

Disagreements between reviewers where one reviewer has recommended a study for inclusion based on full-text review whereas the other has not will automatically be escalated in Covidence. Instances where the third adjudicating reviewer decides yes will be included, whereas those stated no will be excluded. In cases where the 3rd reviewer is uncertain these will be discussed as a group.

Data Extraction

Data will be extracted from studies by two independent reviewers using piloting forms and a standardised extraction tool. Extraction will be structured by the following domains:

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1. Article characteristics, which include title, name of first author, publication year, country of study, study type;
2. Participant details, which include number of participants, demographics, characteristics, study location, recruitment procedures, inclusion and exclusion criteria;
3. Methods, which include identification of AF, definition and application of precision health initiatives, exposure measurement; and,
4. Outcomes, which include measurement tool, timing of measurements, method of aggregation, summary data.

Authors of papers will be contacted to request missing or additional data, where required.

Data Synthesis

Best evidence synthesis will be used. Meta-analysis of intervention (effectiveness) can be carried out; however, we will need to account for substantial heterogeneity between and within included studies. Specifically for our review, outcomes will need to be compared according to the specific health literacy intervention carried out. The process for meta-analysis will be informed following data extraction, based on the assumptions of homogeneity.

If homogeneity is sufficient between groups within these categories, a consensus will be reached between all authors regarding the inclusion within the meta-analysis either as a large group or several subgroups. If eligible studies are clearly homogenous, studies will be allocated to appropriate sub-groups for analysis. This process will be done by each reviewer independently, determining the factors that would allow for the most accurate comparisons to be made.

In the case of substantial heterogeneity between methods and outcomes and/or instances where there are not sufficient number of outcomes that can be analysed (e.g. according to the three questions of 1) knowledge and/or literacy 2) patient action and behaviour change and 3) health outcomes, including quality of life, hospitalisations, morbidity/mortality) we will carry out a best evidence synthesis in the place of meta-analysis.

To assess the impact of heterogeneity across studies, a statistic test of inconsistency will be carried out, returning I^2 which will be reported as a percentage. The Cochrane Handbook for Systematic Reviews of Interventions(13) provides a suggested guide for interpretation of I^2, group percentages into 0-40%, 30-60%, 50-90%, 75-100% to potentially mean no importance, moderate, substantial, and considerable heterogeneity respectively. If present, heterogeneity will be explored, and potential options, such as excluding studies with obvious outlying results, will be explored.

Risk of Bias & Confidence in Cumulative Evidence

Strategies to combat potential publication data will include searches of unpublished studies, contact authors in their respective field, and including grey literature. Further statistical tests, such as the Begg and Mazumbar's rank correlation test and Egger's linear regression model, may be applied to each category and overall analyses. On publication bias being detected, Duval and Tweedie's trim and fill correction may be applied, and the resultant effect sizes and 95% confidence intervals examined in further detail.

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The Downs and Black's 27 item checklist(37) can be used to assess methodological quality of randomised and non-randomised trials. It provides: an overall score for study quality, and a profile of scores for the quality of reporting internal validity, power and external validity.

The data generated from this SR/MA such as Downs and Black's rating of quality and/or risk of bias as well as data extraction will be available as supplementary material along with the final report/publication

A Summary of Findings (SoF) will be created using GRADEpro GDT. The SoF will present the following information where appropriate: absolute risks for the treatment and control, estimates of relative risk, and a ranking of the quality of the evidence based on the risk of bias, directness, heterogeneity, precision and risk of publication bias of the review results. The GRADE(38) approach will be utilised to assess quality of evidence.

Clinical implications

This research aims to comprehensively synthesize through systematic review and meta-analysis the existing published and gray literature of health literacy interventions as a component of precision health to improve a) knowledge and literacy b) patient action and behavior change and c) health outcomes. We anticipate this information being used to inform the assessment and future implementation of health literacy interventions not only improving the use of such terms in research and practice but having impact on patient outcomes, cost-effectiveness and accessibility.

CONTRIBUTORSHIP STATEMENT

All authors have contributed to the focus of this systematic review topic and revisions associated with this protocol manuscript. Specifically, DK, TL, and RM were responsible for the initial conceptualisation of this work. TL, DK, RM led the planning and design, and all authors (TL, DK, KF, NC, MB, RM) were involved in the conduct, reporting and critical review of this manuscript. All authors made substantial contributions to the drafting and critical revision of the work and all authors approved the final manuscript.

COMPETING INTERESTS

There are no competing interests.

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DATA SHARING

Data sharing not applicable as no datasets generated and/or analysed for this study.

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ETHICS AND DISSEMINATION

Approval from a human research ethics committee is not required as human participants will not be involved in this review. We do intend to publish this review on completion in a high-quality peer-reviewed journals, conference presentations and internal staff research meetings.

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APPENDIX

Appendix 1 – Search Strategy

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Logic Grids as of 21/04/2022

MEDLINE (via Pubmed platform)

Atrial Fibrillation	**Health Literacy Initiatives**	**Outcomes**
96,171 hits	*585,959 hits*	*4,482,653 hits*
"atrial fibrillation"[mh] OR "atrial fibrillation"[tiab] OR "auricular fibrillation"[tiab]	"health literacy"[mh] OR "patient medication knowledge"[mh] OR "health communication"[mh] OR "health promotion"[mh] OR "patient education as topic"[mh] OR "health education"[mh] OR "patient participation"[mh] OR "telemedicine"[mh] OR "internet-based intervention"[mh] OR "social media"[mh] OR "mobile applications"[mh] OR "smartphone"[mh] OR "educational technology"[mh] OR "health literacy"[tiab] OR "patient medication knowledge"[tiab] OR "health education"[tiab] OR "health communication*"[tiab] OR "health promotion*"[tiab] OR "patient participation"[tiab] OR "telemedicine"[tiab] OR "telehealth"[tiab] OR "internet-based intervention*"[tiab] OR "social media"[tiab] OR "mobile applications"[tiab] OR "smartphone"[tiab] OR "educational technolog*"[tiab] OR "health literacy interventions"[tiab] OR "education program*"[tiab] OR "educational program*"[tiab] OR "wellness program*"[tiab] OR "health campaign*"[tiab] OR "patient education"[tiab] OR "patient empowerment"[tiab] OR "patient engagement"[tiab] OR "patient activation"[tiab] OR "mobile health"[tiab] OR "mhealth"[tiab] OR "ehealth"[tiab] OR "interactive health literacy"[tiab] OR "web-based intervention*"[tiab] OR "web based intervention*"[tiab] OR "online intervention*"[tiab] OR "internet intervention*"[tiab] OR "mobile app*"[tiab] OR "smartphone app*"[tiab] OR "digital health"[tiab] OR "instructional technolog*"[tiab] OR (("technolog*"[tiab] OR "computer"[tiab] OR "internet"[tiab] OR "online"[tiab]) AND "health literacy"[tiab])	"health knowledge, attitudes, practice"[mh] OR "patient preference"[mh] OR "treatment adherence and compliance"[mh] OR "prognosis"[mh] OR "patient reported outcome measures"[mh] OR "patient outcome assessment"[mh] OR "quality of life"[mh] OR "health status"[mh] OR "activities of daily living"[mh] OR "mortality"[mh] OR "morbidity"[mh] OR "hospitalization"[mh] OR "health knowledge"[tiab] OR "health attitudes"[tiab] OR "health practice"[tiab] OR "patient preference*"[tiab] OR "treatment adherence and compliance"[tiab] OR "prognosis"[tiab] OR "patient reported outcome measure*"[tiab] OR "patient outcome assessment*"[tiab] OR "quality of life"[tiab] OR "health status"[tiab] OR "activities of daily living"[tiab] OR "mortality"[tiab] OR "morbidity"[tiab] OR "hospitalization*"[tiab] OR "hospitalisation*"[tiab] OR "health literacy questionnaire"[tiab] OR "health understanding"[tiab] OR "health competenc*"[tiab] OR "behaviour change"[tiab] OR "treatment adherence"[tiab] OR "therapeutic adherence"[tiab] OR "health-related quality of life"[tiab] OR "HRQOL"[tiab] OR "ADL"[tiab] OR "functional capacity"[tiab] OR "stroke risk"[tiab] OR "atrial fibrillation complication*"[tiab] OR "atrial fibrillation burden"[tiab]
		647 combined hits

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Logic Grids as of 21/04/2022
EMBASE (via OVID platform)

Atrial Fibrillation	Health Literacy Initiatives	Outcomes
172,878 hits	*640,603 hits*	*4,596,175 hits*
exp "atrial fibrillation"/ or "atrial fibrillation".ti,ab.	"health literacy"/ or "patient education"/ or "health education"/ or "medical information"/ or "health promotion"/ or "patient participation"/ or "patient engagement"/ or "telemedicine"/ or "telehealth"/ or "web-based intervention"/ or "social media"/ or exp "mobile application"/ or exp "smartphone"/ or "educational technology"/ or "health literacy".ti,ab. or "patient education".ti,ab. or "health education".ti,ab. or "medical information".ti,ab. or "health promotion".ti,ab. or "patient participation".ti,ab. or "patient engagement".ti,ab. or "telemedicine".ti,ab. or "telehealth".ti,ab. or "web?based intervention*".ti,ab. or "social media".ti,ab. or "mobile app*".ti,ab. or "smartphone".ti,ab. or "educational technolog*".ti,ab. or "patient medication knowledge".ti,ab. or "health communication*".ti,ab. or "internet-based intervention*".ti,ab. or "health literacy interventions".ti,ab. or "education* program*".ti,ab. or "wellness program*".ti,ab. or "health campaign*".ti,ab. or "patient empowerment".ti,ab. or "patient activation".ti,ab. or "mobile health".ti,ab. or "mhealth".ti,ab. or "ehealth".ti,ab. or "interactive health literacy".ti,ab. or "online intervention*".ti,ab. or "internet intervention*".ti,ab. or "smartphone app*".ti,ab. or "digital health".ti,ab. or "instructional technolog*".ti,ab. or (("technolog*" or "computer" or "internet" or "online").ti,ab. AND "health literacy".ti,ab.)	"patient preference"/ or exp "patient compliance"/ or "prognosis"/ or "patient-reported outcome"/ or "outcome assessment"/ or exp "quality of life"/ or "health status"/ or "fitness"/ or "functional status"/ or "health disparity"/ or "physical mobility"/ or "daily life activity"/ or "mortality"/ or "morbidity"/ or "hospitalization/" or "behaviour change"/ or "patient preference*".ti,ab. or "patient compliance".ti,ab. or "prognosis".ti,ab. or "patient-reported outcome*".ti,ab. or "outcome assessment".ti,ab. or "quality of life".ti,ab. or "health status".ti,ab. or "fitness".ti,ab. or "functional status".ti,ab. or "health disparity".ti,ab. or "physical mobility".ti,ab. or "daily life activity".ti,ab. or "activities of daily living".ti,ab. or "mortality".ti,ab. or "morbidity".ti,ab. or "hospitali?ation".ti,ab. or "behaviour change".ti,ab. or "health knowledge".ti,ab. or "health attitudes".ti,ab. or "health practice".ti,ab. or "health literacy questionnaire".ti,ab. or "health understanding".ti,ab. or "health competenc*".ti,ab. or "treatment adherence".ti,ab. or "therapeutic adherence".ti,ab. or "health-related quality of life".ti,ab. or "HRQOL".ti,ab. or "ADL".ti,ab. or "stroke risk".ti,ab. or "atrial fibrillation complication*".ti,ab. or "atrial fibrillation burden".ti,ab.
		1,141 combined hits

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Logic Grids as of 21/04/2022

Web of Science (via Clarivate platform)

Atrial Fibrillation	**Health Literacy Initiatives**	**Outcomes**
114,231 hits	*405,908 hits*	*2,449,416 hits*
TS=("atrial fibrillation")	TS=("health literacy" OR "patient medication knowledge" OR "health communication" OR "health promotion" OR "patient education" OR "health education" OR "patient participation" OR "telemedicine" OR "internet-based intervention" OR "social media" OR "mobile applications" OR "smartphone" OR "educational technology" OR "health communication*" OR "health promotion*" OR "health literacy interventions" OR "education* program*" OR "wellness program*" OR "health campaign*" OR "patient education" OR "patient empowerment" OR "patient engagement" OR "patient activation" OR "mobile health" OR "mhealth" OR "ehealth" OR "interactive health literacy" OR "web-based intervention*" OR "web based intervention*" OR "online intervention*" OR "internet intervention*" OR "mobile app*" OR "smartphone app*" OR "digital health" OR "instructional technolog*" OR (("technolog*" OR "computer" OR "internet" OR "online") AND "health literacy"))	TS=("health knowledge" OR "health attitudes" OR "health practice" OR "patient preference" OR "treatment adherence" OR "treatment compliance" OR "prognosis" OR "patient reported outcome measures" OR "patient outcome assessment" OR "quality of life" OR "health status" OR "activities of daily living" OR "mortality" OR "morbidity" OR "hospitali?ation*" OR "health literacy questionnaire" OR "health understanding" OR "health competenc*" OR "behaviour change" OR "treatment adherence" OR "therapeutic adherence" OR "health-related quality of life" OR "HRQOL" OR "ADL" OR "functional capacity" OR "stroke risk" OR "atrial fibrillation complication*" OR "atrial fibrillation burden")
		319 combined hits

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Logic Grids as of 21/04/2022
CINAHL (via EBSCOhost platform)

Atrial Fibrillation	**Health Literacy Initiatives**	**Outcomes**
39,320 hits	*298,598 hits*	*1,150,418 hits*
MH "atrial fibrillation" OR TI "atrial fibrillation" OR AB "atrial fibrillation"	MH "health literacy" OR MH "health promotion" OR MH "health education" OR MH "patient education" OR MH "telemedicine" OR MH "telehealth" OR MH "internet-based intervention" OR MH "social media+" OR MH "mobile applications" OR MH "smartphone" OR MH "educational technology" OR TI ("health literacy" OR "health promotion*" OR "health education" OR "patient education" OR "telemedicine" OR "telehealth" OR "internet-based intervention" OR "social media" OR "mobile app*" OR "smartphone" OR "educational technolog*" OR "patient medication knowledge" OR "health communication" OR "patient participation" OR "health literacy interventions" OR "educational program*" OR "education program*" OR "wellness program*" OR "health campaign*" OR "patient empowerment" OR "patient engagement" OR "patient activation" OR "mobile health" OR "mhealth" OR "ehealth" OR "smartphone app*" OR "interactive health literacy" OR "web-based intervention*" OR "web based intervention*" OR "online intervention*" OR "internet intervention*" OR "digital health" OR "instructional technolog*" OR (("technolog*" OR "computer" OR "internet" OR "online") AND "health literacy"))) OR AB ("health literacy" OR "health promotion*" OR "health education" OR "patient education" OR "telemedicine" OR "telehealth" OR "internet-based intervention" OR "social media" OR "mobile app*" OR "smartphone" OR "educational technolog*" OR "patient medication knowledge" OR "health communication" OR "patient participation" OR "health literacy interventions" OR "educational program*" OR "education program*" OR "wellness program*" OR "health campaign*" OR "patient empowerment" OR "patient engagement" OR "patient activation" OR "mobile health" OR "mhealth" OR "ehealth" OR "smartphone app*" OR "interactive health literacy" OR "web-based intervention*" OR "web based intervention*" OR "online intervention*" OR "internet intervention*" OR "digital health" OR "instructional technolog*" OR (("technolog*" OR "computer" OR "internet" OR "online") AND "health literacy")))	MH "health knowledge" OR MH "patient preference" OR MH "prognosis" OR MH "treatment outcomes" OR MH "outcome assessment" OR MH "patient-reported outcomes" OR MH "quality of life" OR MH "health status+" OR MH "activities of daily living" OR MH "mortality" OR MH "morbidity" OR MH "hospitalization" OR MH "behavioural changes" OR MH "functional status" OR TI ("health knowledge" OR "health practice" OR "patient preference" OR "prognosis" OR "treatment outcomes" OR "outcome assessment" OR "patient-reported outcomes" OR "quality of life" OR "health status" OR "activities of daily living" OR "mortality" OR "morbidity" OR "hospitali#ation" OR "behavioural changes" OR "functional status" OR "treatment adherence" OR "therapeutic adherence" OR "health literacy questionnaire" OR "health understanding" OR "health competenc*" OR "health-related quality of life" OR "HRQOL" OR "ADL" OR "stroke risk" OR "atrial fibrillation complication*" OR "atrial fibrillation burden") OR AB ("health knowledge" OR "health practice" OR "patient preference" OR "prognosis" OR "treatment outcomes" OR "outcome assessment" OR "patient-reported outcomes" OR "quality of life" OR "health status" OR "activities of daily living" OR "mortality" OR "morbidity" OR "hospitali#ation" OR "behavioural changes" OR "functional status" OR "treatment adherence" OR "therapeutic adherence" OR "health literacy questionnaire" OR "health understanding" OR "health competenc*" OR "health-related quality of life" OR "HRQOL" OR "ADL" OR "stroke risk" OR "atrial fibrillation complication*" OR "atrial fibrillation burden")
		232 combined hits

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Logic Grids as of 21/04/2022

Emcare (via OVID platform)

Atrial Fibrillation	**Health Literacy Initiatives**	**Outcomes**
41,599 hits	*276,531 hits*	*1,210,113 hits*
exp "atrial fibrillation"/ or "atrial fibrillation".ti,ab.	"health literacy"/ or "patient education"/ or "health education"/ or "medical information"/ or "health promotion"/ or "patient participation"/ or "patient engagement"/ or "telemedicine"/ or "telehealth"/ or "web-based intervention"/ or "social media"/ or exp "mobile application"/ or exp "smartphone"/ or "educational technology"/ or "health literacy".ti,ab. or "patient education".ti,ab. or "health education".ti,ab. or "medical information".ti,ab. or "health promotion".ti,ab. or "patient participation".ti,ab. or "patient engagement".ti,ab. or "telemedicine".ti,ab. or "telehealth".ti,ab. or "web?based intervention*".ti,ab. or "social media".ti,ab. or "mobile app*".ti,ab. or "smartphone".ti,ab. or "educational technolog*".ti,ab. or "patient medication knowledge".ti,ab. or "health communication*".ti,ab. or "internet-based intervention*".ti,ab. or "health literacy interventions".ti,ab. or "education* program*".ti,ab. or "wellness program*".ti,ab. or "health campaign*".ti,ab. or "patient empowerment".ti,ab. or "patient activation".ti,ab. or "mobile health".ti,ab. or "mhealth".ti,ab. or "ehealth".ti,ab. or "interactive health literacy".ti,ab. or "online intervention*".ti,ab. or "internet intervention*".ti,ab. or "smartphone app*".ti,ab. or "digital health".ti,ab. or "instructional technolog*".ti,ab. or (("technolog*" or "computer" or "internet" or "online").ti,ab. AND "health literacy".ti,ab.)	"patient preference"/ or exp "patient compliance"/ or "prognosis"/ or "patient-reported outcome"/ or "outcome assessment"/ or exp "quality of life"/ or "health status"/ or "fitness"/ or "functional status"/ or "health disparity"/ or "physical mobility"/ or "daily life activity"/ or "mortality"/ or "morbidity"/ or "hospitalization/" or "behaviour change"/ or "patient preference*".ti,ab. or "patient compliance".ti,ab. or "prognosis".ti,ab. or "patient-reported outcome*".ti,ab. or "outcome assessment".ti,ab. or "quality of life".ti,ab. or "health status".ti,ab. or "fitness".ti,ab. or "functional status".ti,ab. or "health disparity".ti,ab. or "physical mobility".ti,ab. or "daily life activity".ti,ab. or "activities of daily living".ti,ab. or "mortality".ti,ab. or "morbidity".ti,ab. or "hospitali?ation".ti,ab. or "behaviour change".ti,ab. or "health knowledge".ti,ab. or "health attitudes".ti,ab. or "health practice".ti,ab. or "health literacy questionnaire".ti,ab. or "health understanding".ti,ab. or "health competenc*".ti,ab. or "treatment adherence".ti,ab. or "therapeutic adherence".ti,ab. or "health-related quality of life".ti,ab. or "HRQOL".ti,ab. or "ADL".ti,ab. or "stroke risk".ti,ab. or "atrial fibrillation complication*".ti,ab. or "atrial fibrillation burden".ti,ab.
		337 combined hits

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Logic Grids as of 21/04/2022
Cochrane Library

Atrial Fibrillation	Health Literacy Initiatives	Outcomes
13,040 hits	*63,570 hits*	*428,751 hits*
[mh "atrial fibrillation"] OR "atrial fibrillation":ti,ab OR "auricular fibrillation":ti,ab	[mh "health literacy"] OR [mh "patient medication knowledge"] OR [mh "health communication"] OR [mh "health promotion"] OR [mh "patient education as topic"] OR [mh "health education"] OR [mh "patient participation"] OR [mh "telemedicine"] OR [mh "internet-based intervention"] OR [mh "social media"] OR [mh "mobile applications"] OR [mh "smartphone"] OR [mh "educational technology"] OR "health literacy":ti,ab OR "patient medication knowledge":ti,ab OR "health education":ti,ab OR (health NEXT communication*):ti,ab OR (health NEXT promotion*):ti,ab OR "patient participation":ti,ab OR "telemedicine":ti,ab OR "telehealth":ti,ab OR (internet-based NEXT intervention*):ti,ab OR "social media":ti,ab OR "mobile applications":ti,ab OR "smartphone":ti,ab OR (educational NEXT technolog*):ti,ab OR "health literacy interventions":ti,ab OR (education NEXT program*):ti,ab OR (educational program*):ti,ab OR (wellness NEXT program*):ti,ab OR (health NEXT campaign*):ti,ab OR "patient education":ti,ab OR "patient empowerment":ti,ab OR "patient engagement":ti,ab OR "patient activation":ti,ab OR "mobile health":ti,ab OR "mhealth":ti,ab OR "ehealth":ti,ab OR "interactive health literacy":ti,ab OR (web-based NEXT intervention*):ti,ab OR (web NEXT based NEXT intervention*):ti,ab OR (online NEXT intervention*):ti,ab OR (internet NEXT intervention*):ti,ab OR (mobile NEXT app*):ti,ab OR (smartphone NEXT app*):ti,ab OR "digital health":ti,ab OR (instructional NEXT technolog*):ti,ab OR ((technolog*:ti,ab OR computer:ti,ab OR internet:ti,ab OR online:ti,ab) AND "health literacy":ti,ab)	[mh "health knowledge, attitudes, practice"] OR [mh "patient preference"] OR [mh "treatment adherence and compliance"] OR [mh "prognosis"] OR [mh "patient reported outcome measures"] OR [mh "patient outcome assessment"] OR [mh "quality of life"] OR [mh "health status"] OR [mh "activities of daily living"] OR [mh "mortality"] OR [mh "morbidity"] OR [mh "hospitalization"] OR "health knowledge":ti,ab OR "health attitudes":ti,ab OR "health practice":ti,ab OR "patient preference*":ti,ab OR "treatment adherence and compliance":ti,ab OR "prognosis":ti,ab OR (patient NEXT reported NEXT outcome NEXT measure*):ti,ab OR (patient NEXT outcome NEXT assessment*):ti,ab OR "quality of life":ti,ab OR "health status":ti,ab OR "activities of daily living":ti,ab OR "mortality":ti,ab OR "morbidity":ti,ab OR (hospitalization*):ti,ab OR (hospitalisation*):ti,ab OR "health literacy questionnaire":ti,ab OR "health understanding":ti,ab OR (health NEXT competenc*):ti,ab OR "behaviour change":ti,ab OR "treatment adherence":ti,ab OR "therapeutic adherence":ti,ab OR "health-related quality of life":ti,ab OR "HRQOL":ti,ab OR "ADL":ti,ab OR "functional capacity":ti,ab OR "stroke risk":ti,ab OR "atrial fibrillation complication*":ti,ab OR "atrial fibrillation burden":ti,ab
		174 combined hits

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Location in Manuscript
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	Page 1: Title, Page 2: Abstract
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	Page 4: Abstract (CRD42022304835)
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Page 1: Title Page
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Page 1: Title Page, Page 8: Contributorship Statement
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	Page 8, Financial Disclosures
Sponsor	5b	Provide name for the review funder and/or sponsor	N/A
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	N/A
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Page 3
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Page 4
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting,	Page 4-6

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		time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Page 6
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Appendix 1
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Page 6
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	Page 6
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Page 7
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	Page 7
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Page 5-6
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Page 7-8
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	Page 7
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication	Page 7

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		bias across studies, selective reporting within studies)	
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	Page 8

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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BMJ Open

Impact of health literacy and its interventions on health outcomes in those with atrial fibrillation: a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2022-065407.R2
Article Type:	Protocol
Date Submitted by the Author:	09-Nov-2022
Complete List of Authors:	Lathlean, Timothy; The University of Adelaide Adelaide Medical School, Faculty of Health and Medical Sciences; South Australian Health and Medical Research Institute Limited, Kieu, Don; The University of Adelaide Adelaide Medical School, Faculty of Health and Medical Sciences Franke, Kyle; The University of Adelaide Adelaide Medical School, Faculty of Health and Medical Sciences O'Callaghan, Nathan; Commonwealth Scientific and Industrial Research Organisation, Precision Health Future Science Platform; South Australian Health and Medical Research Institute Limited Boyd, Mark A; The University of Adelaide Adelaide Medical School, Faculty of Health and Medical Sciences; Northern Adelaide Local Health Network, Lyell McEwin Hospital Mahajan, Rajiv; The University of Adelaide Faculty of Health and Medical Sciences, Adelaide Medical School; Northern Adelaide Local Health Network, Lyell McEwin Hospital
Primary Subject Heading:	Cardiovascular medicine
Secondary Subject Heading:	Patient-centred medicine
Keywords:	Health informatics < BIOTECHNOLOGY & BIOINFORMATICS, Adult cardiology < CARDIOLOGY, PREVENTIVE MEDICINE

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Manuscripts

Impact of health literacy and its interventions on health outcomes in those with atrial fibrillation: a systematic review protocol

AUTHORS AND AFFILIATIONS

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CONFLICT OF INTEREST DISCLOSURES:

The University of Adelaide reports receiving on behalf of Dr Mahajan lecture and/or consulting fees from Abbott, Medtronic, Bayer, Biotronik, and Pfizer. The University of Adelaide reports receiving on behalf of Dr Mahajan research funding from Abbott, Bayer, and Medtronic. The remaining authors do not report any conflicts of interest.

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Keywords: Atrial fibrillation, precision health, health literacy, health outcomes, quality of life

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ABSTRACT

Introduction

Atrial fibrillation (AF) is associated with increased risk of stroke, heart failure and death. Health literacy, an aspect that falls within precision health, has been recognised as an important factor. We will be focusing on the impact of these interventions specifically to AF and its health outcomes.

Methods and Analysis

This protocol is informed by the Preferred Reporting Items for Systematic review and Meta-Analysis Protocols (PRISMA-P). The results will be reported in line with the Preferred Reporting Items for Systematic review and Meta-Analysis (PRISMA) to determine the impacts of health literacy interventions on atrial fibrillation outcomes. Searches will be carried out on databases including MEDLINE, EMBASE, Web of Science, CINAHL, Emcare, Cochrane library, and Google Scholar. Citations will be collected via Endnote 20, then into Covidence for duplicate removal, and article screening. Extraction will occur using a standardised extraction tool and studies will be synthesised using best evidence synthesis. Downs and Black's checklist will be used for risk of bias and assessment of overall quality of evidence will utilise the Grading of Recommendations, Assessment, Development, and Evaluation approach (GRADE).

Ethics and Dissemination

Approval from human research ethics committee is not required. Dissemination will occur in peer-reviewed journals and conference presentations.

STRENGTHS AND LIMITATIONS OF STUDY

- This will be the first review to operationalise health literacy interventions in people with atrial fibrillation.
- The protocol for this systematic review and meta-analysis follows the PRISMA-P guidelines.
- This systematic review will use the Downs and Black checklist for quality appraisal and the GRADE recommendations for certainty of evidence.
- The meta-analysis will provide a synthesis of the evidence of health literacy interventions consistent with a) knowledge b) action and c) health outcomes.
- Different study designs and definitions of health literacy interventions as well as varied outcomes may limit the ability the extent of comparative analysis.

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INTRODUCTION

Background & Rationale

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia and represents significant morbidity and mortality worldwide.(1) The worldwide prevalence of AF is estimated at 0.51% of population and projected to increase further.(2) In addition to an increased risk of stroke, heart failure, and all-cause death, AF continues to contribute to the growing burden on the healthcare system and its resources.(1)

Health literacy plays a critical role within a patient's healthcare and quality of life, and improving it has been a recognised goal by the World Health Organisation (WHO).(3, 4) Health literacy has been defined according to three types: functional health literacy, interactive health literacy and critical health literacy. (5) It involves people's knowledge and interactions with health information via four dimensions: accessing, understanding, appraising, and applying. (6) Health literacy can be considered as an aspect of 'precision health', which involve the processes of lifetime monitoring and collection of health data, analysis of collected data, and subsequent informed, personalised decision making and interventions.(7-10) A recent scoping review (9) identified nine categories of data that can be collected: sociodemographic, anthropometric, clinical, genomic, perspective, behavioural, psychological, nutritional, and socioenvironmental data.

Currently, a range of definitions and terms have been used in the health literacy literature, with different aims and contexts, which has created uncertainty and a lack of clarity as to what these constitute. This is particularly pronounced when considering health literacy interventions as a core component of precision health initiatives. Our systematic review and meta-analysis aims to evaluate the effectiveness of health literacy interventions for people with AF. In doing so, we aim to synthesise the current literature to identify the impacts of health literacy interventions on a) health literacy and knowledge; b) patient action and behaviour change arising from health literacy and knowledge and c) health outcomes (e.g. quality of life, hospitalisations and/or morbidity/mortality) as a consequence to action taken arising from improvements in health literacy.

Objectives

This systematic review will explore precision health, in particular, the impact of health literacy and its interventions, on atrial fibrillation outcomes. Thus, the systematic review will endeavour to answer the following:

1. How do health literacy interventions generate additional benefits for precision health initiatives?
2. What questionnaires and tools are currently available and have been used to assess health literacy interventions generally and in AF?
3. How effective are health literacy interventions in improving both knowledge and subsequent beneficial actions for AF?
4. How effective are health literacy interventions that improve knowledge and subsequent beneficial actions in resulting in improved health outcomes such as quality of life, hospitalisations and reduced morbidity/mortality?

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METHODS

Guidelines and Registration

This protocol for our systematic review was done with guidance and adherence to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 guidelines,(11, 12), Cochrane guidelines, (13) and other reviews already conducted in this area.(14, 15) Registration was submitted to the International Prospective Register of Systematic Reviews (PROSPERO), an international database for systematic reviews prospectively registered by the Centre for Reviews and Dissemination of the University of York (https://www.crd.york.ac.uk/prospero). This occurred on [29th April 2022] and was last updated on [10th May 2022] (registration number [CRD42022304835]).

Patient and Public Involvement

None

Eligibility Criteria

Selection of studies will consider the following:

Study Designs

Randomised control trials (RCTs) and observational studies will be included. Conference presentations, thesis, secondary research (review articles) will be excluded.

Population

Included studies that have enrolled individuals with a primary diagnosis of non-valvular AF. Any age, baseline physical activity, or different backgrounds will be considered, as well as those in various settings, such as hospital-based, hospital outpatient clinicals, private clinics, primary care, research-based specialist university clinics. No restrictions will be placed on duration of symptoms nor time since diagnosis. Studies in which participants are diagnosed with valvular AF will be excluded.

Interventions

This review will consider studies that evaluate interventions and initiatives focused on health literacy that are aimed at improving any of the following 1) knowledge and/or literacy 2) patient action and behaviour change and 3) health outcomes, including quality of life, hospitalisations, morbidity/mortality.

Interventions may endeavour to improve any dimension,(6) typology,(5) or similar aspect of health literacy, such as a patient's ability to read, write, approach numeracy, communicate, and/or use electronic technology.(16)

Interventions considered can address health literacy in any domain(6) or setting, and may be in the form of different medium. This can involve face-to-face mediums, such as verbal communication, written communication, visual aids, and supportive systems.(16, 17). We also acknowledge the growing usage of remote mediums and digital initiatives, such as online resources, web-based resources, social media, and mobile applications.

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Such intervention fits within Gambhir et al.'s(7) model of precision health within the "Integrated health portal", allowing for learning and adaptation that feeds back into the rest of the model.

Comparator

In addition to usual clinical care, we will be considering behavioural data, sociodemographic data, and clinical data as part of our comparators. Viana et al.'s(9) scoping review on precision health research showed that these were the top three types of data used for precision health (behavioural (48%), sociodemographic (44%), clinical (43.6%)).

Outcomes

Three categories of outcomes will be collected: outcomes concerned with the knowledge and understanding of AF following intervention; outcomes measuring any resulting patient action following intervention; and finally, outcomes measuring health, wellness, and quality of life. Due to the broad nature of measures for outcomes, the following is not an exhaustive list. Any additional outcomes suitable for the three categories listed above will be considered and agreed upon by all researchers involved.

Examples of outcomes include

- Those related to knowledge and understanding
    - Test of Functional Health Literacy in Adults (TOFHLA)(18)/short-TOFHLA (s-TOFHLA)(19)
    - Rapid Estimate of Adult Literacy in Medicine (REALM)(20)/REALM-Short Form (REALM-SF)(21)
    - Medical achievement reading test (MART)(22)
- Those related to action
    - Self-reported measures
- Those related to health, wellness, and quality of life
    - All-cause mortality
    - Hospitalisations, and re-hospitalisation rates
    - Heart function
    - Quality of life(23)
        - Generic instruments – Medical Outcome Study Short-Form 36 (SF-36),(24) 12-Item Short-Form Health Survey (SF-12),(25) EuroQoL 5-Dimensional (EQ-5D)(26, 27)
        - AF-specific instruments – Atrial Fibrillation Effect on QualiTy-of-Life (AFEQT),(28) AF-QOL-18,(29) QLAF questionnaire,(30) Atrial Fibrillation Quality of Life Questionnaire (AFQLQ)(31)
    - Symptom scales(23)
    - Physical activity – Internal Physical Activity Questionnaire (IPAQ),(32) Lawton-Brody iADL scale,(33) Katz-ADL scale(34)
    - Self-reported measures
        - Ability to perform ADLs
        - Work and productivity

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Information Sources

Searches will be carried out on MEDLINE (via PubMed platform), EMBASE (via OVID platform), Web of Science (via Clarivate platform), CINAHL (via EBSCOhost platform), Emcare (via OVID platform), Cochrane Library, and Google Scholar.(35)

Additionally, to ensure completeness, gray literature(36) will be included and citation searching of included articles will be undertaken. Gray literature, or evidence not published in commercial publications can include academic papers, such as theses/dissertations, research and/or governmental reports, conference papers. We intend to search for such literature through resources such as OpenGrey, Scopus, and Proquest Digital Dissertations and Digital Dissertations and Conference Papers Index (CPI). We also aim to hand-search through relevant conference proceedings and abstracts such as the Cardiac Society of Australia and New Zealand (CSANZ), Asia Pacific Heart Rhythm Society (APHRS) and Heart Rhythm Society (HRS) Conferences.

Search Strategy

The draft search strategy was developed initially using the PubMed platform, medical subject headings (MeSH) and key phrases identified during preliminary searches. These were initially performed in January 2022 and then updated in April 2022 and assisted in refining search terms within the PICO structure. Phrases specific to the intervention, comparator, and outcome components were developed by first identifying core key phrases relevant to our questions, then searching PubMed using these phrases and expanding on synonyms using existing articles and search strategies. We based our search on previous literature including Viana et al.'s(9) review of precision health. The search will be limited to 2012 onwards as well as English language to minimise risk to including studies published before the research field was mature.

The draft search strategy was reviewed by a librarian specialising in medicine and was also reviewed by the research team. A complete search strategy, which has been adapted to syntax and subject headings of other databases, is included in Appendix 1.

Data Management & Selection Process

Citations from databases will be imported to EndNote 20 (Clarivate Analytics) for article collection, then to Covidence for the remaining processes. Duplicates will be removed electronically via Covidence and checked manually. Eligibility for inclusion will involve two independent reviewers, and firstly will be done based on title and abstract screening, followed by full-text review. At both stages, reviewers will be blinded to each other's decisions until discussions occur.

Disagreements between reviewers where one reviewer has recommended a study for inclusion based on full-text review whereas the other has not will automatically be escalated in Covidence. Instances where the third adjudicating reviewer decides yes will be included, whereas those stated no will be excluded. In cases where the 3rd reviewer is uncertain these will be discussed as a group.

Data Extraction

Data will be extracted from studies by two independent reviewers using piloting forms and a standardised extraction tool. Extraction will be structured by the following domains:

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1. Article characteristics, which include title, name of first author, publication year, country of study, study type;
2. Participant details, which include number of participants, demographics, characteristics, study location, recruitment procedures, inclusion and exclusion criteria;
3. Methods, which include identification of AF, definition and application of precision health initiatives, exposure measurement; and,
4. Outcomes, which include measurement tool, timing of measurements, method of aggregation, summary data.

Authors of papers will be contacted to request missing or additional data, where required.

Data Synthesis

Best evidence synthesis will be used. Meta-analysis of intervention (effectiveness) can be carried out; however, we will need to account for substantial heterogeneity between and within included studies. Specifically for our review, outcomes will need to be compared according to the specific health literacy intervention carried out. The process for meta-analysis will be informed following data extraction, based on the assumptions of homogeneity.

If homogeneity is sufficient between groups within these categories, a consensus will be reached between all authors regarding the inclusion within the meta-analysis either as a large group or several subgroups. If eligible studies are clearly homogenous, studies will be allocated to appropriate sub-groups for analysis. This process will be done by each reviewer independently, determining the factors that would allow for the most accurate comparisons to be made.

In the case of substantial heterogeneity between methods and outcomes and/or instances where there are not sufficient number of outcomes that can be analysed (e.g. according to the three questions of 1) knowledge and/or literacy 2) patient action and behaviour change and 3) health outcomes, including quality of life, hospitalisations, morbidity/mortality) we will carry out a best evidence synthesis in the place of meta-analysis.

To assess the impact of heterogeneity across studies, a statistic test of inconsistency will be carried out, returning I^2 which will be reported as a percentage. The Cochrane Handbook for Systematic Reviews of Interventions(13) provides a suggested guide for interpretation of I^2, group percentages into 0-40%, 30-60%, 50-90%, 75-100% to potentially mean no importance, moderate, substantial, and considerable heterogeneity respectively. If present, heterogeneity will be explored, and potential options, such as excluding studies with obvious outlying results, will be explored.

Risk of Bias & Confidence in Cumulative Evidence

Strategies to combat potential publication data will include searches of unpublished studies, contact authors in their respective field, and including grey literature. Further statistical tests, such as the Begg and Mazumbar's rank correlation test and Egger's linear regression model, may be applied to each category and overall analyses. On publication bias being detected, Duval and Tweedie's trim and fill correction may be applied, and the resultant effect sizes and 95% confidence intervals examined in further detail.

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The Downs and Black's 27 item checklist(37) can be used to assess methodological quality of randomised and non-randomised trials. It provides: an overall score for study quality, and a profile of scores for the quality of reporting internal validity, power and external validity.

The data generated from this SR/MA such as Downs and Black's rating of quality and/or risk of bias as well as data extraction will be available as supplementary material along with the final report/publication

A Summary of Findings (SoF) will be created using GRADEpro GDT. The SoF will present the following information where appropriate: absolute risks for the treatment and control, estimates of relative risk, and a ranking of the quality of the evidence based on the risk of bias, directness, heterogeneity, precision and risk of publication bias of the review results. The GRADE(38) approach will be utilised to assess quality of evidence.

Clinical implications

This research aims to comprehensively synthesize through systematic review and meta-analysis the existing published and gray literature of health literacy interventions as a component of precision health to improve a) knowledge and literacy b) patient action and behavior change and c) health outcomes. We anticipate this information being used to inform the assessment and future implementation of health literacy interventions not only improving the use of such terms in research and practice but having impact on patient outcomes, cost-effectiveness and accessibility.

CONTRIBUTORSHIP STATEMENT

All authors (TL, DK, KBF, NO, MAB, RM) have contributed to the focus of this systematic review topic and revisions associated with this protocol manuscript. Specifically, DK, TL, and RM were responsible for the initial conceptualisation of this work. TL, DK, RM led the planning and design, and all authors were involved in the conduct and reporting of this manuscript. All authors (TL, DK, KBF, NO, MAB, RM) made substantial contributions to the drafting and critical revision of the work and all authors (TL, DK, KBF, NO, MAB, RM) approved the final manuscript.

COMPETING INTERESTS

There are no competing interests.

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DATA SHARING

Data sharing not applicable as no datasets generated and/or analysed for this study.

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ETHICS AND DISSEMINATION

Approval from a human research ethics committee is not required as human participants will not be involved in this review. We do intend to publish this review on completion in a high-quality peer-reviewed journals, conference presentations and internal staff research meetings.

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APPENDIX

Appendix 1 – Search Strategy

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Logic Grids as of 21/04/2022

MEDLINE (via Pubmed platform)

Atrial Fibrillation	**Health Literacy Initiatives**	**Outcomes**
96,171 hits	*585,959 hits*	*4,482,653 hits*
"atrial fibrillation"[mh] OR "atrial fibrillation"[tiab] OR "auricular fibrillation"[tiab]	"health literacy"[mh] OR "patient medication knowledge"[mh] OR "health communication"[mh] OR "health promotion"[mh] OR "patient education as topic"[mh] OR "health education"[mh] OR "patient participation"[mh] OR "telemedicine"[mh] OR "internet-based intervention"[mh] OR "social media"[mh] OR "mobile applications"[mh] OR "smartphone"[mh] OR "educational technology"[mh] OR "health literacy"[tiab] OR "patient medication knowledge"[tiab] OR "health education"[tiab] OR "health communication*"[tiab] OR "health promotion*"[tiab] OR "patient participation"[tiab] OR "telemedicine"[tiab] OR "telehealth"[tiab] OR "internet-based intervention*"[tiab] OR "social media"[tiab] OR "mobile applications"[tiab] OR "smartphone"[tiab] OR "educational technolog*"[tiab] OR "health literacy interventions"[tiab] OR "education program*"[tiab] OR "educational program*"[tiab] OR "wellness program*"[tiab] OR "health campaign*"[tiab] OR "patient education"[tiab] OR "patient empowerment"[tiab] OR "patient engagement"[tiab] OR "patient activation"[tiab] OR "mobile health"[tiab] OR "mhealth"[tiab] OR "ehealth"[tiab] OR "interactive health literacy"[tiab] OR "web-based intervention*"[tiab] OR "web based intervention*"[tiab] OR "online intervention*"[tiab] OR "internet intervention*"[tiab] OR "mobile app*"[tiab] OR "smartphone app*"[tiab] OR "digital health"[tiab] OR "instructional technolog*"[tiab] OR (("technolog*"[tiab] OR "computer"[tiab] OR "internet"[tiab] OR "online"[tiab]) AND "health literacy"[tiab])	"health knowledge, attitudes, practice"[mh] OR "patient preference"[mh] OR "treatment adherence and compliance"[mh] OR "prognosis"[mh] OR "patient reported outcome measures"[mh] OR "patient outcome assessment"[mh] OR "quality of life"[mh] OR "health status"[mh] OR "activities of daily living"[mh] OR "mortality"[mh] OR "morbidity"[mh] OR "hospitalization"[mh] OR "health knowledge"[tiab] OR "health attitudes"[tiab] OR "health practice"[tiab] OR "patient preference*"[tiab] OR "treatment adherence and compliance"[tiab] OR "prognosis"[tiab] OR "patient reported outcome measure*"[tiab] OR "patient outcome assessment*"[tiab] OR "quality of life"[tiab] OR "health status"[tiab] OR "activities of daily living"[tiab] OR "mortality"[tiab] OR "morbidity"[tiab] OR "hospitalization*"[tiab] OR "hospitalisation*"[tiab] OR "health literacy questionnaire"[tiab] OR "health understanding"[tiab] OR "health competenc*"[tiab] OR "behaviour change"[tiab] OR "treatment adherence"[tiab] OR "therapeutic adherence"[tiab] OR "health-related quality of life"[tiab] OR "HRQOL"[tiab] OR "ADL"[tiab] OR "functional capacity"[tiab] OR "stroke risk"[tiab] OR "atrial fibrillation complication*"[tiab] OR "atrial fibrillation burden"[tiab]
		647 combined hits

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Logic Grids as of 21/04/2022
EMBASE (via OVID platform)

Atrial Fibrillation	Health Literacy Initiatives	Outcomes
172,878 hits	*640,603 hits*	*4,596,175 hits*
exp "atrial fibrillation"/ or "atrial fibrillation".ti,ab.	"health literacy"/ or "patient education"/ or "health education"/ or "medical information"/ or "health promotion"/ or "patient participation"/ or "patient engagement"/ or "telemedicine"/ or "telehealth"/ or "web-based intervention"/ or "social media"/ or exp "mobile application"/ or exp "smartphone"/ or "educational technology"/ or "health literacy".ti,ab. or "patient education".ti,ab. or "health education".ti,ab. or "medical information".ti,ab. or "health promotion".ti,ab. or "patient participation".ti,ab. or "patient engagement".ti,ab. or "telemedicine".ti,ab. or "telehealth".ti,ab. or "web?based intervention*".ti,ab. or "social media".ti,ab. or "mobile app*".ti,ab. or "smartphone".ti,ab. or "educational technolog*".ti,ab. or "patient medication knowledge".ti,ab. or "health communication*".ti,ab. or "internet-based intervention*".ti,ab. or "health literacy interventions".ti,ab. or "education* program*".ti,ab. or "wellness program*".ti,ab. or "health campaign*".ti,ab. or "patient empowerment".ti,ab. or "patient activation".ti,ab. or "mobile health".ti,ab. or "mhealth".ti,ab. or "ehealth".ti,ab. or "interactive health literacy".ti,ab. or "online intervention*".ti,ab. or "internet intervention*".ti,ab. or "smartphone app*".ti,ab. or "digital health".ti,ab. or "instructional technolog*".ti,ab. or (("technolog*" or "computer" or "internet" or "online").ti,ab. AND "health literacy".ti,ab.)	"patient preference"/ or exp "patient compliance"/ or "prognosis"/ or "patient-reported outcome"/ or "outcome assessment"/ or exp "quality of life"/ or "health status"/ or "fitness"/ or "functional status"/ or "health disparity"/ or "physical mobility"/ or "daily life activity"/ or "mortality"/ or "morbidity"/ or "hospitalization/" or "behaviour change"/ or "patient preference*".ti,ab. or "patient compliance".ti,ab. or "prognosis".ti,ab. or "patient-reported outcome*".ti,ab. or "outcome assessment".ti,ab. or "quality of life".ti,ab. or "health status".ti,ab. or "fitness".ti,ab. or "functional status".ti,ab. or "health disparity".ti,ab. or "physical mobility".ti,ab. or "daily life activity".ti,ab. or "activities of daily living".ti,ab. or "mortality".ti,ab. or "morbidity".ti,ab. or "hospitali?ation".ti,ab. or "behaviour change".ti,ab. or "health knowledge".ti,ab. or "health attitudes".ti,ab. or "health practice".ti,ab. or "health literacy questionnaire".ti,ab. or "health understanding".ti,ab. or "health competenc*".ti,ab. or "treatment adherence".ti,ab. or "therapeutic adherence".ti,ab. or "health-related quality of life".ti,ab. or "HRQOL".ti,ab. or "ADL".ti,ab. or "stroke risk".ti,ab. or "atrial fibrillation complication*".ti,ab. or "atrial fibrillation burden".ti,ab.
		1,141 combined hits

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Logic Grids as of 21/04/2022

Web of Science (via Clarivate platform)

Atrial Fibrillation	**Health Literacy Initiatives**	**Outcomes**
114,231 hits	*405,908 hits*	*2,449,416 hits*
TS=("atrial fibrillation")	TS=("health literacy" OR "patient medication knowledge" OR "health communication" OR "health promotion" OR "patient education" OR "health education" OR "patient participation" OR "telemedicine" OR "internet-based intervention" OR "social media" OR "mobile applications" OR "smartphone" OR "educational technology" OR "health communication*" OR "health promotion*" OR "health literacy interventions" OR "education* program*" OR "wellness program*" OR "health campaign*" OR "patient education" OR "patient empowerment" OR "patient engagement" OR "patient activation" OR "mobile health" OR "mhealth" OR "ehealth" OR "interactive health literacy" OR "web-based intervention*" OR "web based intervention*" OR "online intervention*" OR "internet intervention*" OR "mobile app*" OR "smartphone app*" OR "digital health" OR "instructional technolog*" OR (("technolog*" OR "computer" OR "internet" OR "online") AND "health literacy"))	TS=("health knowledge" OR "health attitudes" OR "health practice" OR "patient preference" OR "treatment adherence" OR "treatment compliance" OR "prognosis" OR "patient reported outcome measures" OR "patient outcome assessment" OR "quality of life" OR "health status" OR "activities of daily living" OR "mortality" OR "morbidity" OR "hospitali?ation*" OR "health literacy questionnaire" OR "health understanding" OR "health competenc*" OR "behaviour change" OR "treatment adherence" OR "therapeutic adherence" OR "health-related quality of life" OR "HRQOL" OR "ADL" OR "functional capacity" OR "stroke risk" OR "atrial fibrillation complication*" OR "atrial fibrillation burden")
		319 combined hits

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Logic Grids as of 21/04/2022
CINAHL (via EBSCOhost platform)

Atrial Fibrillation	**Health Literacy Initiatives**	**Outcomes**
39,320 hits	*298,598 hits*	*1,150,418 hits*
MH "atrial fibrillation" OR TI "atrial fibrillation" OR AB "atrial fibrillation"	MH "health literacy" OR MH "health promotion" OR MH "health education" OR MH "patient education" OR MH "telemedicine" OR MH "telehealth" OR MH "internet-based intervention" OR MH "social media+" OR MH "mobile applications" OR MH "smartphone" OR MH "educational technology" OR TI ("health literacy" OR "health promotion*" OR "health education" OR "patient education" OR "telemedicine" OR "telehealth" OR "internet-based intervention" OR "social media" OR "mobile app*" OR "smartphone" OR "educational technolog*" OR "patient medication knowledge" OR "health communication" OR "patient participation" OR "health literacy interventions" OR "educational program*" OR "education program*" OR "wellness program*" OR "health campaign*" OR "patient empowerment" OR "patient engagement" OR "patient activation" OR "mobile health" OR "mhealth" OR "ehealth" OR "smartphone app*" OR "interactive health literacy" OR "web-based intervention*" OR "web based intervention*" OR "online intervention*" OR "internet intervention*" OR "digital health" OR "instructional technolog*" OR (("technolog*" OR "computer" OR "internet" OR "online") AND "health literacy"))) OR AB ("health literacy" OR "health promotion*" OR "health education" OR "patient education" OR "telemedicine" OR "telehealth" OR "internet-based intervention" OR "social media" OR "mobile app*" OR "smartphone" OR "educational technolog*" OR "patient medication knowledge" OR "health communication" OR "patient participation" OR "health literacy interventions" OR "educational program*" OR "education program*" OR "wellness program*" OR "health campaign*" OR "patient empowerment" OR "patient engagement" OR "patient activation" OR "mobile health" OR "mhealth" OR "ehealth" OR "smartphone app*" OR "interactive health literacy" OR "web-based intervention*" OR "web based intervention*" OR "online intervention*" OR "internet intervention*" OR "digital health" OR "instructional technolog*" OR (("technolog*" OR "computer" OR "internet" OR "online") AND "health literacy")))	MH "health knowledge" OR MH "patient preference" OR MH "prognosis" OR MH "treatment outcomes" OR MH "outcome assessment" OR MH "patient-reported outcomes" OR MH "quality of life" OR MH "health status+" OR MH "activities of daily living" OR MH "mortality" OR MH "morbidity" OR MH "hospitalization" OR MH "behavioural changes" OR MH "functional status" OR TI ("health knowledge" OR "health practice" OR "patient preference" OR "prognosis" OR "treatment outcomes" OR "outcome assessment" OR "patient-reported outcomes" OR "quality of life" OR "health status" OR "activities of daily living" OR "mortality" OR "morbidity" OR "hospitali#ation" OR "behavioural changes" OR "functional status" OR "treatment adherence" OR "therapeutic adherence" OR "health literacy questionnaire" OR "health understanding" OR "health competenc*" OR "health-related quality of life" OR "HRQOL" OR "ADL" OR "stroke risk" OR "atrial fibrillation complication*" OR "atrial fibrillation burden") OR AB ("health knowledge" OR "health practice" OR "patient preference" OR "prognosis" OR "treatment outcomes" OR "outcome assessment" OR "patient-reported outcomes" OR "quality of life" OR "health status" OR "activities of daily living" OR "mortality" OR "morbidity" OR "hospitali#ation" OR "behavioural changes" OR "functional status" OR "treatment adherence" OR "therapeutic adherence" OR "health literacy questionnaire" OR "health understanding" OR "health competenc*" OR "health-related quality of life" OR "HRQOL" OR "ADL" OR "stroke risk" OR "atrial fibrillation complication*" OR "atrial fibrillation burden")
		232 combined hits

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Logic Grids as of 21/04/2022

Emcare (via OVID platform)

Atrial Fibrillation	**Health Literacy Initiatives**	**Outcomes**
41,599 hits	*276,531 hits*	*1,210,113 hits*
exp "atrial fibrillation"/ or "atrial fibrillation".ti,ab.	"health literacy"/ or "patient education"/ or "health education"/ or "medical information"/ or "health promotion"/ or "patient participation"/ or "patient engagement"/ or "telemedicine"/ or "telehealth"/ or "web-based intervention"/ or "social media"/ or exp "mobile application"/ or exp "smartphone"/ or "educational technology"/ or "health literacy".ti,ab. or "patient education".ti,ab. or "health education".ti,ab. or "medical information".ti,ab. or "health promotion".ti,ab. or "patient participation".ti,ab. or "patient engagement".ti,ab. or "telemedicine".ti,ab. or "telehealth".ti,ab. or "web?based intervention*".ti,ab. or "social media".ti,ab. or "mobile app*".ti,ab. or "smartphone".ti,ab. or "educational technolog*".ti,ab. or "patient medication knowledge".ti,ab. or "health communication*".ti,ab. or "internet-based intervention*".ti,ab. or "health literacy interventions".ti,ab. or "education* program*".ti,ab. or "wellness program*".ti,ab. or "health campaign*".ti,ab. or "patient empowerment".ti,ab. or "patient activation".ti,ab. or "mobile health".ti,ab. or "mhealth".ti,ab. or "ehealth".ti,ab. or "interactive health literacy".ti,ab. or "online intervention*".ti,ab. or "internet intervention*".ti,ab. or "smartphone app*".ti,ab. or "digital health".ti,ab. or "instructional technolog*".ti,ab. or (("technolog*" or "computer" or "internet" or "online").ti,ab. AND "health literacy".ti,ab.)	"patient preference"/ or exp "patient compliance"/ or "prognosis"/ or "patient-reported outcome"/ or "outcome assessment"/ or exp "quality of life"/ or "health status"/ or "fitness"/ or "functional status"/ or "health disparity"/ or "physical mobility"/ or "daily life activity"/ or "mortality"/ or "morbidity"/ or "hospitalization/" or "behaviour change"/ or "patient preference*".ti,ab. or "patient compliance".ti,ab. or "prognosis".ti,ab. or "patient-reported outcome*".ti,ab. or "outcome assessment".ti,ab. or "quality of life".ti,ab. or "health status".ti,ab. or "fitness".ti,ab. or "functional status".ti,ab. or "health disparity".ti,ab. or "physical mobility".ti,ab. or "daily life activity".ti,ab. or "activities of daily living".ti,ab. or "mortality".ti,ab. or "morbidity".ti,ab. or "hospitali?ation".ti,ab. or "behaviour change".ti,ab. or "health knowledge".ti,ab. or "health attitudes".ti,ab. or "health practice".ti,ab. or "health literacy questionnaire".ti,ab. or "health understanding".ti,ab. or "health competenc*".ti,ab. or "treatment adherence".ti,ab. or "therapeutic adherence".ti,ab. or "health-related quality of life".ti,ab. or "HRQOL".ti,ab. or "ADL".ti,ab. or "stroke risk".ti,ab. or "atrial fibrillation complication*".ti,ab. or "atrial fibrillation burden".ti,ab.
		337 combined hits

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Logic Grids as of 21/04/2022
Cochrane Library

Atrial Fibrillation	Health Literacy Initiatives	Outcomes
13,040 hits	*63,570 hits*	*428,751 hits*
[mh "atrial fibrillation"] OR "atrial fibrillation":ti,ab OR "auricular fibrillation":ti,ab	[mh "health literacy"] OR [mh "patient medication knowledge"] OR [mh "health communication"] OR [mh "health promotion"] OR [mh "patient education as topic"] OR [mh "health education"] OR [mh "patient participation"] OR [mh "telemedicine"] OR [mh "internet-based intervention"] OR [mh "social media"] OR [mh "mobile applications"] OR [mh "smartphone"] OR [mh "educational technology"] OR "health literacy":ti,ab OR "patient medication knowledge":ti,ab OR "health education":ti,ab OR (health NEXT communication*):ti,ab OR (health NEXT promotion*):ti,ab OR "patient participation":ti,ab OR "telemedicine":ti,ab OR "telehealth":ti,ab OR (internet-based NEXT intervention*):ti,ab OR "social media":ti,ab OR "mobile applications":ti,ab OR "smartphone":ti,ab OR (educational NEXT technolog*):ti,ab OR "health literacy interventions":ti,ab OR (education NEXT program*):ti,ab OR (educational program*):ti,ab OR (wellness NEXT program*):ti,ab OR (health NEXT campaign*):ti,ab OR "patient education":ti,ab OR "patient empowerment":ti,ab OR "patient engagement":ti,ab OR "patient activation":ti,ab OR "mobile health":ti,ab OR "mhealth":ti,ab OR "ehealth":ti,ab OR "interactive health literacy":ti,ab OR (web-based NEXT intervention*):ti,ab OR (web NEXT based NEXT intervention*):ti,ab OR (online NEXT intervention*):ti,ab OR (internet NEXT intervention*):ti,ab OR (mobile NEXT app*):ti,ab OR (smartphone NEXT app*):ti,ab OR "digital health":ti,ab OR (instructional NEXT technolog*):ti,ab OR ((technolog*:ti,ab OR computer:ti,ab OR internet:ti,ab OR online:ti,ab) AND "health literacy":ti,ab)	[mh "health knowledge, attitudes, practice"] OR [mh "patient preference"] OR [mh "treatment adherence and compliance"] OR [mh "prognosis"] OR [mh "patient reported outcome measures"] OR [mh "patient outcome assessment"] OR [mh "quality of life"] OR [mh "health status"] OR [mh "activities of daily living"] OR [mh "mortality"] OR [mh "morbidity"] OR [mh "hospitalization"] OR "health knowledge":ti,ab OR "health attitudes":ti,ab OR "health practice":ti,ab OR "patient preference*":ti,ab OR "treatment adherence and compliance":ti,ab OR "prognosis":ti,ab OR (patient NEXT reported NEXT outcome NEXT measure*):ti,ab OR (patient NEXT outcome NEXT assessment*):ti,ab OR "quality of life":ti,ab OR "health status":ti,ab OR "activities of daily living":ti,ab OR "mortality":ti,ab OR "morbidity":ti,ab OR (hospitalization*):ti,ab OR (hospitalisation*):ti,ab OR "health literacy questionnaire":ti,ab OR "health understanding":ti,ab OR (health NEXT competenc*):ti,ab OR "behaviour change":ti,ab OR "treatment adherence":ti,ab OR "therapeutic adherence":ti,ab OR "health-related quality of life":ti,ab OR "HRQOL":ti,ab OR "ADL":ti,ab OR "functional capacity":ti,ab OR "stroke risk":ti,ab OR "atrial fibrillation complication*":ti,ab OR "atrial fibrillation burden":ti,ab
		174 combined hits

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Location in Manuscript
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	Page 1: Title, Page 2: Abstract
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	Page 4: Abstract (CRD42022304835)
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Page 1: Title Page
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Page 1: Title Page, Page 8: Contributorship Statement
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	Page 8, Financial Disclosures
Sponsor	5b	Provide name for the review funder and/or sponsor	N/A
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	N/A
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Page 3
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Page 4
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting,	Page 4-6

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		time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Page 6
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Appendix 1
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Page 6
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	Page 6
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Page 7
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	Page 7
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Page 5-6
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Page 7-8
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	Page 7
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication	Page 7

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		bias across studies, selective reporting within studies)	
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	Page 8

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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